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ScienceDirect
Precambrian Research 156 (2007) 125–152
Precambrian Research
www.elsevier.com/locate/precamres

Circum-Iapetus paleogeography of the Precambrian–Cambrian transition with a new paleomagnetic constraint from Laurentia

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Received 27 July 2006; received in revised form 10 March 2007; accepted 19 March 2007

Abstract

The paleogeography of the Precambrian–Cambrian transition is still poorly known, but is fundamental for understanding Late Neoproterozoic climate extremes and the spatial associations of the rapidly evolving Ediacaran and Cambrian fauna. This period has also been proposed to host unusual geodynamic events such as rapid plate motions and/or the bulk tumbling of the Earth with respect to its spin axis, called true polar wander (TPW). New paleomagnetic and ^{40}Ar–^{39}Ar geochronologic results have been obtained from a pair of shallowly emplaced syenitic intrusions in western Quebec. The Mont Rigaud and the Chatham-Grenville stocks are related to a failed rift arm of the Iapetus Ocean. Both intrusions bear hornblendes which provide overlapping ^{40}Ar–^{39}Ar plateau ages of 533.2 ± 1.1 Ma (Mont Rigaud) and 531.4 ± 3.4 Ma (Chatham-Grenville), corresponding to Early Cambrian (Nemakit-Daldynian) time. Characteristic paleomagnetic (reversed) directions from both intrusions are easterly and shallow ($D = 89.3°$, $I = 30.1°$; $\alpha_{95} = 6.2$, $k = 38.8$; $N = 15$ sites), and the Mont Rigaud intrusion also carries antipodal normal-polarity directions. The combined result yields a north paleopole at location 11.9°S 184.6°E; $dp = 3.8°$ $dm = 6.9°$, placing the Montreal region at $16 \pm 4°$S paleolatitude in the Early Cambrian, in good agreement with Late Ediacaran and Middle to Late Cambrian paleomagnetic results from Laurentia. Laurentia likely resided at low southerly paleolatitudes throughout the Cambrian, ruling out a major inertial-interchange TPW event for Early to Middle Cambrian time. A comparison of Late Ediacaran to Middle Cambrian paleomagnetic results from Laurentia, Baltica, Siberia and elements of West Gondwana implies that by Early Cambrian time a large paleolatitudinal gap existed between Laurentia and its presumed conjugate rift margins, thus placing a wide Iapetus Ocean between them. The position of Baltica within the circum-Iapetus paleogeography is contentious, due to differing interpretations of the few paleomagnetic results. During Ediacaran time, Laurentia underwent large apparent polar wander that is manifested as motion from low latitude at 615 Ma to the south polar region at 590–575 Ma, and then rapidly back to low latitude by 565–550 Ma. The large and perhaps rapid apparent polar wander of Laurentia and other continents in the Ediacaran remains enigmatic and controversial.
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Keywords: Paleomagnetism; Iapetus Ocean; Baltica; Early Cambrian; Ediacaran loop; True polar wander; ^{40}Ar–^{39}Ar geochronology; Hornblende

1. Introduction

Late Neoproterozoic to Early Cambrian time is of singular significance in Earth history, being the period of a major diversification of life, and of severe extremes in climate (Grotzinger et al., 1995; Hoffman and Schrag, 2002). This period features widespread rifting and orogenic activity that accompanied the breakup of the supercontinent Rodinia, the opening of a new ocean, Iapetus, and the assembly of the supercontinent Gondwana (Hoffman, 1991; Dalziel, 1997).

Paleogeography prior to the Middle Cambrian is poorly known, however, with few well determined, well-

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0301-9268/$ – see front matter © 2007 Elsevier B.V. All rights reserved.
doi:10.1016/j.precamres.2007.03.004

dated paleomagnetic constraints (Buchan et al., 2000; Meert and Powell, 2001). Part of the difficulty lies in the absence, prior to the onset of Cambrian shelly fauna, of useful biostratigraphic index fossils for constraining the age of remanence-bearing rocks, thus forcing a greater reliance on radiometric dating constraints. A more particular problem for studies in the terminal Neoproterozoic–Cambrian transition has been the unclear age of the Precambrian–Cambrian boundary itself, now known to be 543 Ma (Gradstein et al., 2004). As recently as a decade ago, estimates of this singularly important boundary had ranged from 650 to 570 Ma (summarized in Bowring and Erwin, 1998), leading to radiometric dates of 570–543 Ma Late Neoproterozoic (Ediacaran Period; Knoll et al., 2004) age being erroneously assigned to the Early Cambrian, and to similar correlation errors within the Cambrian. As a result of this confusion, geological and paleomagnetic data have often been incorrectly related, leading to poor constraints upon paleogeography. Published paleomagnetic results in Ediacaran–Cambrian time can now be reassessed within the constraints of the improved time scale.

Existing paleomagnetic data imply rapid latitudinal motions for Laurentia from the south pole to near the equator between 570 and 550 Ma (Cawood et al., 2001). Into Cambrian time, Gondwana underwent its final assembly and its West Africa portion moved to the south polar region between 550 and 510 Ma (Meert and Van der Voo, 1996). These rapid apparent motions complicate attempts to reconstruct the paleogeography, and have led to non-uniformitarian proposals such as rapid plate motion during the terminal Neoproterozoic opening of the Iapetus (McCausland and Hodych, 1998; Cawood et al., 2001) or the contribution of substantial true polar wander (TPW) during the Early Cambrian (Kirschvink et al., 1997; Evans, 1998).

This study provides new hornblende and potassium feldspar ^{40}Ar–^{39}Ar age dating and primary paleomagnetic results for the Mont Rigaud and Chatham-Grenville stocks of western Quebec. The paleomagnetic results apply to Laurentia for Early Cambrian time, a period which is not well represented for Laurentia. The ^{40}Ar–^{39}Ar ages also extend the region of recognized Early Cambrian rift-related magmatism of Laurentia's Iapetan margin to include the northeastern Laurentia, confirming that rift-related activity took place there for ca. 80 myears during Ediacaran to Early Cambrian time.

2. Geological setting

In Ediacaran to Cambrian time eastern Laurentia experienced a major rifting and magmatic event that presaged the development of its early Paleozoic Iapetan passive margin (Kumarapeli and Saull, 1966; Kumarapeli, 1993). The rift activity included the Ottawa-Bonnechere aulacogen, a failed rift arm which extends into the Canadian Shield along the present-day St. Lawrence and Ottawa River valleys (Fig. 1).

The rift-related magmatism in eastern Laurentia is present in two age pulses, a volumetrically minor ca. 750 Ma largely felsic group in the south and central Appalachians and a major latest Neoproterozoic pulse in the central to northeastern Appalachians (Aleinikoff et al., 1995). The younger pulse of Late Neoproterozoic rift-related magmatism along the incipient Iapetan margin (Fig. 1) occurred over a protracted period and exhibits a range of magmatic compositions, from the tholeittic 615 Ma Long Range dykes, to bimodal rift volcanism of the ca. 570 Ma Catoctin Formation and the final, more distinctly alkalic volcanism of the 554 Ma Tibbit Hill and 550 Ma Skinner Cove Formations (Cawood et al., 2001; Puffer, 2002).

The evolution in magmatic setting and composition has been interpreted to represent possibly plume-influenced continental rifting between Laurentia and an adjacent continent, with incipient rifting having occurred at ca. 615 Ma followed by late-stage alkalic volcanism prior to the Early Cambrian onset of Iapetan seafloor-spreading (Kumarapeli, 1993; Bingen et al., 1998; Puffer, 2002). An earliest Cambrian rift-to-drift transition is inferred from Cambro-Ordovician subsidence curves that are derived from the passive margin successions of western Newfoundland and the central Appalachians (Bond et al., 1984). A widespread carbonate platform and archaeocyathan reefs of Atdabanian age in the Forteau Formation of southeastern Labrador further indicate that a warm-water passive margin had been established by mid-Early Cambrian time (Williams and Hiscott, 1987).

For this work, two rift-related intrusions have been sampled for paleomagnetic and ^{40}Ar–^{39}Ar geochronologic study. The Mont Rigaud and Chatham-Grenville stocks are a pair of silica-saturated syenitic–granitic intrusions west of Montreal (Figs. 1 and 2). The intrusions are emplaced in the craton adjacent to or within the Ottawa-Bonnechere aulacogen, and have not undergone significant post-emplacement tilting (Philpotts, 1976; Malka et al., 2000). Mont Rigaud forms a prominent hill within the Ottawa valley overlooking the town of Rigaud, whereas the Chatham-Grenville stock forms part of the cratonic hinterland to the north of the ancient rift escarpment (Fig. 2). Both intrusions exhibit succes-

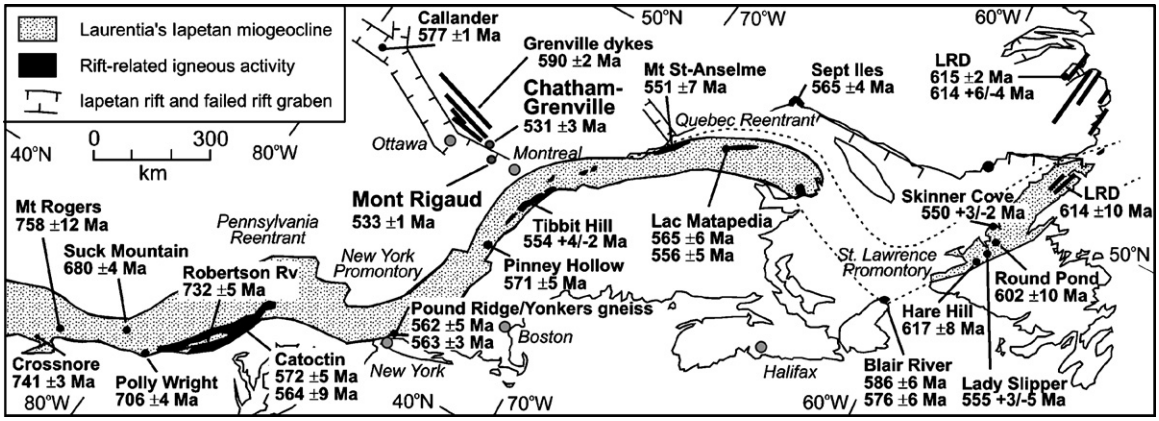


Fig. 1. Distribution of Neoproterozoic rift-related magmatic rocks along the incipient Iapetan margin of Laurentia, in present-day eastern North America (after Cawood et al., 2001). LRD refers to the Long Range dykes of southeastern Labrador and northern Newfoundland. Unit ages are drawn from the following references: Blair River mategabbro dykes – Miller and Barr (2004); Callander complex – Kamo et al. (1995); Catoctin Formation metavolcanics and dykes – Aleinikoff et al. (1995); Chatham-Grenville stock – this study; Crossnore pluton – Su et al. (1994); Grenville dykes – Kamo et al. (1995); Hare Hill granite – van Berkel and Currie (1988); Lac Matapedia volcanics – Cox and Hodych (2005); Lady Slipper pluton – Cawood et al. (1996); Long Range dykes – Kamo et al. (1989), Kamo and Gower (1994), Stukas and Reynolds (1974); Mont Rigaud stock – this study, see also Malka et al. (2000); Mount Rogers rhyolite – Aleinikoff et al. (1995); Mt. St-Anselme volcanics – Cox and Hodych (2005); Pinney Hollow Formation metafelsite – Walsh and Aleinikoff (1999); Polly Wright Cove pluton – Tollo et al. (2004); Robertson River batholith – Lukert and Banks (1984); Round Pond granite – Williams et al. (1985); Sept Iles layered intrusion – Higgins and van Breemen (1998); Suck Mountain pluton – Tollo et al. (2004); Tibbit Hill Formation metavolcanics – Kumarapeli et al. (1989); Pound Ridge granite gneiss and Yonkers gneiss – Tollo et al. (2004). Age errors age quoted as 2σ.

sive intrusive phases which reflect a simple progression to increasing silica content, from feldspar-hornblende syenite to true granite (Greig, 1968; Higgins, 1985), consistent with fractional crystallization and minor crustal assimilation during a single magmatic event (Malka et al., 2000; Higgins, 1985).

2.1. Mont Rigaud stock

Malka et al. (2000) divide the Mont Rigaud stock into three intrusive phases: unit 1 forms the core of the intrusion and consists of dark-coloured feldspar porphyry-hornblende syenite, unit 2 is a light grey horn-

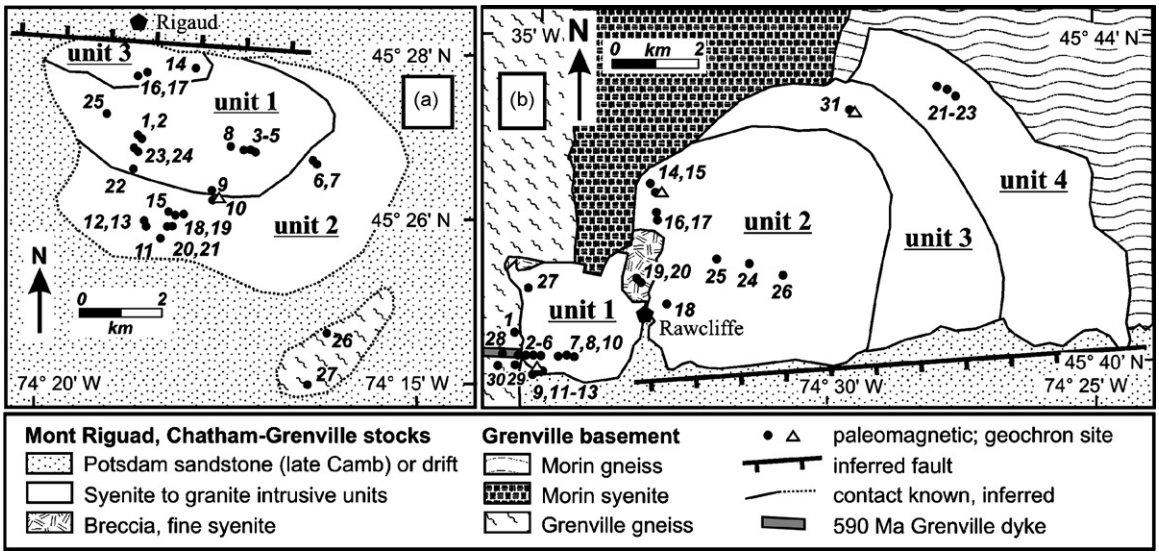


Fig. 2. Geology and sampling sites for: (a) the Mont Rigaud stock, modified from Malka et al. (2000); (b) the Chatham-Grenville stock, modified from Higgins (1985).

blende syenite that outcrops extensively to the east and south of unit 1, and the dominantly granitic unit 3 intrudes the northwestern part of unit 1 near the town of Rigaud (Fig. 2a). All three units are cut by minor aplite and dark syenitic dykes (Greig, 1968). Geochemical and Nd isotopic data indicate that the three intrusive units are co-magmatic, and are most simply interpreted as representing fractional crystallization of an initial syenitic magma during a short period of time (Malka et al., 2000). The presence of miarolitic cavities and chilled contacts between intrusive phases (Greig, 1968; Malka et al., 2000) suggests that the Mont Rigaud syenite was emplaced at a shallow crustal level.

The Mont Rigaud stock is surrounded by flat-lying beds of the Late Cambrian Potsdam Group quartz arenite sandstone (Dix et al., 2004) which was deposited on Grenville gneiss of the Frontenac-Adirondack Belt (Easton, 2000). The gneiss is exposed locally as an inlier 2 km to the southeast of the intrusion (Fig. 2a). No intrusive contact between the stock and the basement gneiss is exposed, nor is there a demonstrated relationship between the stock and the surrounding Potsdam Group sandstone. The north margin of the Mont Rigaud stock exhibits a steep cliff face that likely represents an east-west trending fault within the aulocogen.

An imprecise whole-rock K–Ar age of 450 Ma has previously been determined for the Mont Rigaud stock (Doig and Barton, 1968). More recently, U–Pb zircon ages of $556+15/-12$ and $564+10/-8$ Ma have been obtained from samples of the unit 1 syenite and unit 3 granite, respectively, by the conventional thermal-ionization mass spectrometry (TIMS) method (Malka et al., 2000). The U–Pb ages imply that the stock crystallized at ca. 560 Ma, but none of the analysed zircon fractions were concordant, and the maximum $^{206}Pb/^{238}U$ fraction age is ~540 Ma. A sample from the unit 2 chill margin was collected in this study for ^{40}Ar–^{39}Ar hornblende and potassium feldspar geochronology.

2.2. Chatham-Grenville stock

The Chatham-Grenville stock exhibits a similar progression from hornblende syenite to true granite in four intrusive phases (Higgins, 1985), and is emplaced in the Morin syenite and older gneiss (Fig. 2b). Unit 1 outcrops in the west as a grey perthitic feldspar porphyry hornblende syenite, containing numerous aplite dykes. Unit 2 lies to the east of unit 1, separated from it along a contact consisting of a dark igneous breccia north of the town of Rawcliffe (Philpotts, 1976). Unit 2 is compositionally variable from grey hornblende syenite to brilliant pink feldspar porphyry quartz syenite. Units 3 and 4 are true granites with variable-to-nil mafic mineral content. Samples from intrusive units 1, 2 and 3 were collected for hornblende and potassium feldspar ^{40}Ar–^{39}Ar geochronology.

Unit 1 syenite of the intrusion cross-cuts members of the east-west trending 590 Ma Grenville dyke swarm (Kamo et al., 1995), providing an opportunity for a baked contact test of the Chatham-Grenville stock on the Grenville dyke. The southern margin of the stock is part of the regional escarpment that marks the ancient faulted margin of the rift. Blue dolostone of the Early Ordovician Beekmantown Group lies horizontally within the Ottawa valley to the south (Philpotts, 1976).

3. ^{40}Ar–^{39}Ar geochronology

Drill core and chip samples from several hornblende-bearing syenite and granite sites in both the Mont Rigaud and Chatham-Grenville stocks were crushed to produce hornblende and potassium feldspar grains for ^{40}Ar–^{39}Ar geochronology (Fig. 2). Hand-picked separates of hornblende and potassium feldspar grains were rinsed with deionized water and dried. Single, non-aggregate grains of 1–2 mm size were then selected under a binocular microscope for further analysis. Several grains from each site were packaged at the University of Michigan Radiogenic Isotope Geochemistry Laboratory in an aluminum foil capsule and irradiated for 300 h at the Phoenix-Ford Memorial Reactor at the University of Michigan. Neutron flux gradients were monitored using the MMhb-1 standard (520.4 Ma; Samson and Alexander, 1987).

During analysis, single grains were step-heated until fusion by an automated argon-ion laser which was defocused to ensure uniform heating. Individual steps consisted of 60 s heating followed by 2 min of gas purification using two 10 l/s SAES getters (ST101 alloy) and a liquid N_2 cold finger to remove the active gasses prior to mass spectrometry. Argon isotopic ratios were measured with a VG1200S mass spectrometer. Extraction line blanks were run after every six steps. Duplicate grains were analysed to assess grain-to-grain variability and the reproducibility of the results. The ^{40}Ar–^{39}Ar plateau ages given below are reported as 1σ and based on the Ar measured during at least five consecutive heating steps that represent >50% of the total gas released, with step ages that pass a chi-squared test of the squared weighted deviations from the weighted average age (McDougall and Harrison, 1999). Plateau ages with 1σ errors are calculated as the inverse variance weighted mean of the ages from the steps which comprise the plateau. Error contribution from decay constant and standard uncertainties have not been included. All argon isotopic data are set

out in an online appendix table at address (*Precambrian Research*) and are summarized in Table 1.

3.1. ^{40}Ar–^{39}Ar results

A sample from the Mont Rigaud site 10 hornblende syenite yielded individual hornblende grains B83a and B83b for analysis. Apparent ages are shown for both grains in Fig. 3a and b as the stepwise cumulative fraction of the total released ^{39}Ar. The hornblendes give replicate plateau ages of 532.0 ± 1.3 Ma (Fig. 3a) and 534.3 ± 1.2 Ma (Fig. 3b) that agree within error and with the total gas age of B83b (532.9 ± 1.3 Ma). The Ar stepwise release shows a clear relationship between

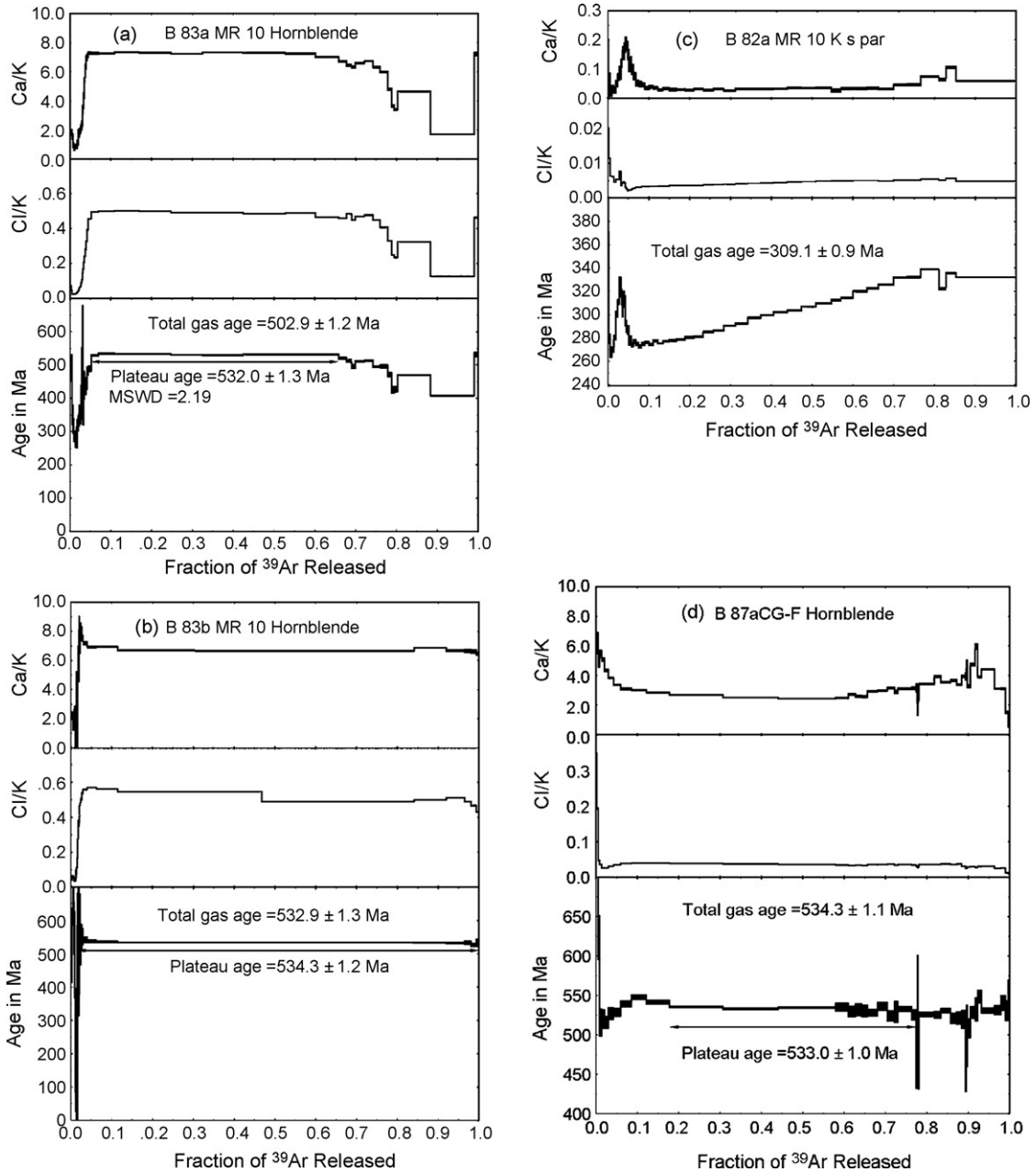


Fig. 3. Argon gas release spectra, plotted as apparent age vs. cumulative ^{39}Ar gas release for hornblende and potassium feldspar grains from site MR10 in the unit 2 chill margin of the Mont Rigaud stock (a–c), and Chatham-Grenville sites CG31 in unit 3 granite ("CG-F" in d,e), CG14 in unit 2 syenite ("CG-H" in f and g), and CG-13 in unit 1 syenite (h–j). Contiguous degassing steps in hornblende that are spanned by an arrowed interval comprise the quoted plateau age at which closure of the K–Ar system is interpreted to have occurred. See text for details.

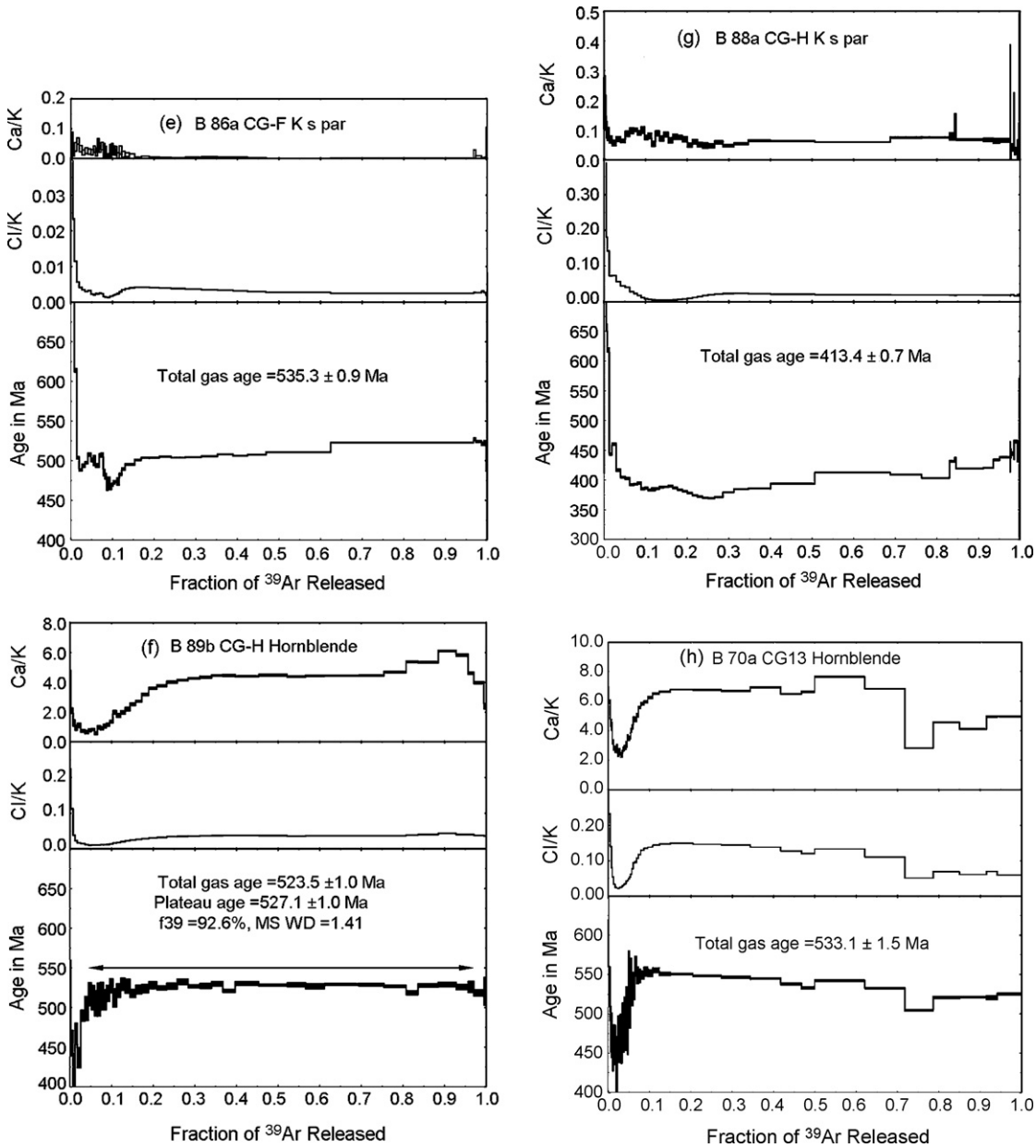


Fig. 3. (*Continued*)

step composition (Cl/K, Ca/K ratios) and the apparent ages for individual steps, indicating that for both grains a single phase is defining the inferred Ar plateau age. Site 10 is sampled from the chill margin of the unit 2 hornblende syenite on unit 1 hornblende feldspar syenite of Mont Rigaud (Fig. 2a). The hornblende ^{40}Ar–^{39}Ar plateau age of 533.2 ± 1.2 Ma (mean of the two replicate grains) therefore likely represents cooling of the intrusion through ~500 °C (McDougall and Harrison, 1999) shortly after its emplacement.

Potassium feldspar B82a from the same site 10 sample shows an increasing apparent age upon degassing that is difficult to interpret (Fig. 3c). The total gas age of ca. 309 Ma is not geologically meaningful, but does imply that the stock did not cool through the closure temperature range of potassium feldspar domains (~280–200 °C; McDougall and Harrison, 1999) in Early Cambrian time, or that the stock was reheated sufficiently at some later time to cause the feldspar to lose radiogenic Ar.

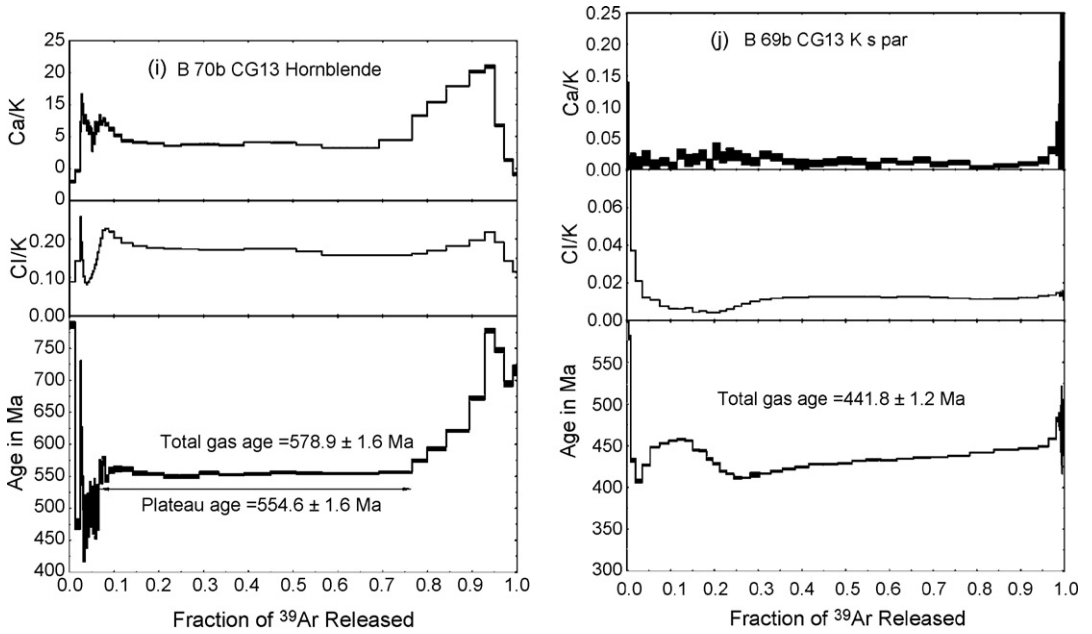


Fig. 3. (*Continued*).

Granite from unit 3 of the Chatham-Grenville stock at site CG31 (CG-F; Fig. 2b) yielded hornblende B87a and potassium feldspar B86a for analysis, along with replicate crystals. An ^{40}Ar–^{39}Ar plateau age of 533.0 ± 1.0 Ma is resolved over 60% of the hornblende degassed Ar in B87a, within error of the grain's total gas age of 534.3 ± 1.1 Ma (Fig. 3d). Replicate hornblende B87b (not shown) yields plateau and total gas ages of 534.1 ± 1.1 Ma and 537.3 ± 1.2 Ma, respectively. Potassium feldspar grains B86a and B86b provide a similar total gas age (535.3 ± 0.9 Ma; Fig. 3e; 536.1 ± 0.9 Ma; not shown) to that of the hornblendes. Progressive step heating of the grains give apparent ages which asymptotically approach ca. 530 Ma, indicating that the feldspars likely have not experienced a thermal event since their initial Early Cambrian formation.

Unit 2 hornblende syenite sampled at site CG14 north of the town of Rawcliffe (CG-H; Fig. 2b) provides hornblende grain B89b (Fig. 3f) which exhibits a plateau age of 527.1 ± 1.0 Ma over the range of steps which show similar composition. Its total gas age of 523.5 ± 1.0 Ma is similar to that of replicate hornblende B89a (526.1 ± 0.9 Ma; not shown). Potassium feldspar B88a (Fig. 3g) does not give a clear estimate of its closure age, having apparent ages of ~370 Ma through ~450 Ma, with a total gas age of ca. 413 Ma.

Feldspar porphyry hornblende syenite of unit 1 sampled at site CG13 in the Scottsdale quarry produced hornblende grains B70a and B70b (Fig. 3h and i), which show differing degassing behaviour. Hornblende B70a has a total gas age of 533.1 ± 1.5 Ma but does not yield a stable plateau. Replicate analysis of grain B70b gives a plateau age of 554.6 ± 1.6 Ma but an anomalously large total gas age of ca. 579 Ma and higher Ca/K and Cl/K ratios than is observed for B70a or other hornblendes from the Chatham-Grenville and Mont Rigaud stocks. The increased apparent ages and the higher Ca/K, Cl/K ratios in the final degassing steps of B70b may indi-

Table 1
Summary of ^{40}Ar–^{39}Ar results shown in Fig. 3 for the Mont Rigaud and Chatham-Grenville stocks, Quebec

Site, unit[a]	Mineral[b]	Apparent plateau age ± 1σ error
MR10, unit 2	hb	532.0 ± 1.3 Ma
MR10, unit 2	hb	534.3 ± 1.2 Ma
MR10, unit 2	K-fsp	(*309.1*) total gas
CG-F, unit 3	hb	533.0 ± 1.0 Ma
CG-F, unit 3	hb	534.1 ± 1.1 Ma
CG-F, unit 3	K-fsp	(*535.3*) total gas
CG-F, unit 3	K-fsp	(*536.1*) total gas
CG-H, unit 2	Hb	(*526.1*) total gas
CG-H, unit 2	Hb	527.1 ± 1.0 Ma
CG-H, unit 2	K-fsp	(*413.4*) total gas
CG13, unit 1	Hb	(*533.1*) total gas
CG13, unit 1	Hb	554.6 ± 1.6 Ma
CG13, unit 1	K-fsp	(*380.9*) total gas
CG13, unit 1	K-fsp	(*441.8*) total gas

[a] Site CG-F is CG31 and site CG-H is CG14 in Fig. 2.
[b] Mineral is: hb, hornblende; K-fsp, potassium feldspar.

cate the presence of pyroxene with trapped excess argon, thus leading to anomalously large ages for the plateau as well as for the final degassing steps. Potassium feldspar B69b (Fig. 3j) has apparent ages of ∼410 Ma climbing to ∼450 Ma in the latter degassing steps, with a total gas age of ca. 442 Ma.

To summarize, hornblende analyses from both intrusions give reproducible plateau ages in three of four sampling sites, with good inter-site agreement. Hornblende grains at site CG13 of the Chatham-Grenville stock either do not yield a stable plateau or appear to be compromised by the degassing of pyroxene with the hornblende, thus leading us to exclude site 13 from the overall age estimate. The remaining three sample sites MR10, CG31 and CG14 (Fig. 2) provide an overall hornblende ^{40}Ar–^{39}Ar age of 531.9 ± 1.4 Ma (weighted mean of the five hornblende plateau ages).

The potassium feldspar age data from all four sample sites record ^{39}Ar loss at some point in their history; all analysed potassium feldspars exhibit an apparent age minima at ∼5–25% degassed ^{39}Ar, followed by a "staircase" degassing spectrum of increasing apparent age. The feldspars B86a and B86b from site CG31 (Fig. 3e) may be considered the least disturbed of those analysed, having the least prominent minimum apparent age (of ca. 460 Ma) and latter step ages that approach the hornblende age from the site, as well as total gas ages of ca. 535 Ma. Overall, the potassium feldspars from the two stocks are most simply interpreted as having initially closed in the early Paleozoic, if not in the Early Cambrian itself, with a subsequent reheating event having caused variable ^{39}Ar loss.

A survey of apatite fission track ages for the Precambrian shield of Ontario (Everitt et al., 2005) shows an anomalously young region which coincides with the Ottawa-Bonnechere rift. Everitt et al. (2005) interpret the anomaly to represent a reheating event in Cretaceous time to at least the apatite annealing temperature of ∼110 °C, possibly related to the passage of the hotspot which created the ca. 115 Ma Monteregian intrusives. The Monteregian event may have heated the Chatham-Grenville and Mont Rigaud stocks sufficiently to partially degas potassium feldspar without significantly affecting the hornblende.

Previously reported imprecise U-Pb zircon concordia intercept ages of 556 + 15/−12 Ma from the Mont Rigaud unit 1 syenite and 564 + 10/−8 Ma from the unit 3 granite (Malka et al., 2000) imply that the stock crystallized in Late Ediacaran time, but as noted above none of the analysed zircon fractions were concordant, and the maximum ^{206}Pb/^{238}U fraction age is ∼540 Ma. Given that the Mont Rigaud magma experienced some crustal contribution (Malka et al., 2000), it is possible that there may be a contribution from older, inherited zircon to the fraction ages. In the absence of concordant zircon analyses the meaning of the imprecise ca. 560 Ma U–Pb zircon ages is unclear.

We interpret the magmatic hornblende and potassium feldspar grain analyses reported here to indicate that the Mont Rigaud and Chatham-Grenville stocks were emplaced at 531.9 ± 1.4 Ma, in Early Cambrian (Nemakit-Daldynian) time, and subsequently were rapidly cooled in their shallow crustal setting.

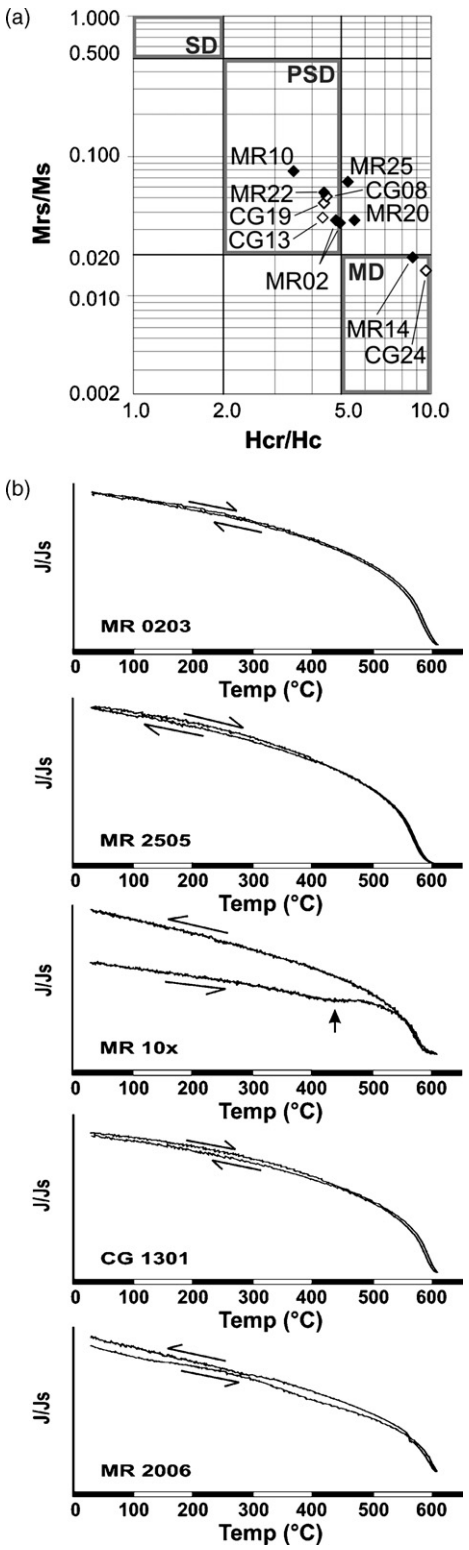
4. Paleomagnetism

The Mont Rigaud and Chatham-Grenville stocks were sampled by portable gas-powered drill and as oriented hand-samples. Both methods produced standard 2.54 cm diameter core samples, and each paleomagnetic site consisted typically of six samples. The Mont Rigaud stock and environs were sampled in 27 sites, providing 320 standard 2.2 cm length specimens for analysis and the Chatham-Grenville stock and environs were sampled in 31 sites, providing 305 specimens.

The specimens underwent thermal and alternating field (AF) stepwise demagnetization at the University of Michigan using a Sapphire Instruments SI-4 AF demagnetizer, an ASC TD-48 thermal demagnetizer, and a 2G cryogenic magnetometer, all set inside a magnetically shielded room which permits entry of only ∼0.2% of the Earth's ambient magnetic field intensity. A subset of 62 specimens from Mont Rigaud were thermally and AF demagnetized at the University of Windsor paleomagnetics laboratory using similar equipment in a similar low ambient magnetic field environment. Results from the Windsor lab agree in all respects with those obtained in Michigan and will henceforth not be distinguished.

4.1. Definition of magnetic mineralogy and the characteristic remanent magnetization (ChRM)

Magnetic hysteresis and backfield parameters (M_{rs}/M_s; H_{cr}/H_c) and thermomagnetic curves were measured at the Institute for Rock Magnetism, University of Minnesota for a suite of samples from both the Mont Rigaud and Chatham-Grenville stocks. Unit 1 samples from both intrusions (Fig. 2) plot in the pseudo-single domain (PSD) magnetite field of the modified Day plot (Dunlop, 2002), whereas unit 2 samples tend to exhibit multidomain (MD) magnetite behaviour (Fig. 4a). Samples from sites MR25 and MR20 are coercively harder than expected for a PSD or MD magnetite phase, suggesting possible contribu-



tions from superparamagnetic magnetite, hematite or pyrrhotite in those samples.

Thermomagnetic heating and cooling curves in a 2.5 T applied field confirm that magnetite is the dominant magnetic mineral in the Mont Rigaud and Chatham-Grenville stocks, as typified by the characteristic Neél temperature for magnetite (∼585 °C) in units 1 and 2 syenite samples (Fig. 4b). All but MR10 have reversible thermomagnetic curves, indicating that no new magnetic minerals were formed during heating. The MR10 sample developed new magnetite probably from titanomagnetite at >450 °C (arrow at inflection point in heating curve; Fig. 4b), suggesting that paleomagnetic analyses need to be restricted to thermal demagnetization steps of 450 °C or less in site MR10. Site MR20 is also from the unit 2 syenite but shows reversibility. In general, thermomagnetic heating curves did not distinguish between units 1 and 2 syenites, or between those sites which carried stable remanence and those which did not (Tables 2a and 2b).

Measurement of the specimens' natural remanent magnetizations (NRM) showed intensities which spanned more than three orders of magnitude, from ∼1 to >8000 mA/m, beyond the measuring range of the cryogenic magnetometer. Units 1 and 2 syenite specimens from both intrusions generally had large NRM intensities (∼100–4000 mA/m; Tables 2a and 2b), whereas unit 3 granite and host gneiss specimens were weaker (1 to ∼100 mA/m). Specimens from the Grenville dykes had very large NRM intensities (Table 2b), and in some cases were measured using a Schonstedt SSM-1 Spinner Magnetometer.

After NRM measurement, two pilot specimens per site were AF demagnetized in 12 steps up to 120 mT. AF step demagnetization typically removed a 'soft' viscous remanent magnetization (VRM) by ∼20 mT. Other pilot specimens were thermally demagnetized in 14 steps up to 580 °C. Most specimens lost between 45 and 95% of their NRM intensity by 20 mT or 450 °C, indicat-

Fig. 4. (a) A modified Day plot (Dunlop, 2002) of the hysteresis parameters H_{cr}/H_c and M_{rs}/M_s for bulk samples from units 1 and 2 of the Mont Rigaud (filled diamond) and Chatham-Grenville (open diamond) intrusions. Unit 1 samples plot in the pseudo-single domain (PSD) magnetite field whereas unit 2 samples tend to exhibit multidomain (MD) magnetite behaviour; (b) thermomagnetic curves for the heating and cooling in a 2.5 T applied field of Mont Rigaud and Chatham-Grenville samples from sites MR02, MR25, MR10, CG13 and MR20. All samples showed a characteristic loss of magnetization between 550 and 590 °C, indicating magnetite to be the dominant magnetic carrier present. The sample from MR10 exhibited the growth of new magnetite at >450 °C (arrow at inflection in heating curve), and was not reversible. See text for discussion.

Table 2a
Site mean remanence data for the 533 Ma Mont Rigaud stock, Quebec (45.45°N, 74.30°W)

Site	Rocktype	Intr. (unit)	NRM (mA/m)	*N*	*e* + *g*	ChRM direction			
						Dec. (°)	Inc. (°)	a_{95} (°)	*k*
MR01	Dark syenite	1	1030	10	5 + 0	103.9	24.8	7.1	115
MR02	Dark syenite	1	1500	12	10 + 0	104.0	22.5	4.9	99
MR03[a]	hb syenite	1	144	11	1 + 0	*65.8*	*43.8*	–	–
MR04	fsp syenite	1	326	9	–				
MR05	fsp syenite	1	132	10	–				
MR06	hb syenite	2	121	8	–				
MR07	hb syenite	2	178	8	–				
MR08	Dark syenite	1	44	8	–				
MR09	Syenite breccia	1	2335	7	5 + 0	80.2	38.2	11.8	42
MR10	Pink syenite	2	170	14	1 + 3	90.9	41.2	10.8	135
MR11	Syenite dyke	2	1940	12	–				
MR12	hb syenite	2	61	11	–				
MR13	hb syenite	2	56	7	–				
MR14	Granite	3	80	13	–				
MR15	hb syenite	2	72	6	–				
MR16	Granite	3	96	5	–				
MR17	Granite	3	69	6	–				
MR18	hb syenite	2	1465	4	–				
MR19	hb syenite	2	1116	4	–				
MR20	hb syenite	2	197	5	–				
MR21	hb syenite	2	54	4	–				
MR22	Purple syenite	1	15	10	3 + 4	277.6	−20.6	14.7	20
MR23	Dyke + syenite	1	580	11	2 + 3	260.0	−18.9	10.5	70
MR24	hb syenite	1	237	8	0 + 4	100.4	36.3	9.4	356
MR25	Dark syenite	1	1900	10	6 + 0	81.1	34.2	6.6	103
MR26[b]	Gneiss	–	187	10	5 + 3	*161.1*	*−80.2*	*18.6*	*10*
6 easterly, reversed sites						93.9	33.3	9.9	46
2 westerly, normal sites						268.8	−20.0	–	–
8 Mont Rigaud sites						**92.5**	**30.0**	**8.8**	**40**

Rocktype: hb, hornblende; fsp, feldspar; Intr. unit, location of site within Mont Rigaud intrusive units 1–3; NRM, mean of specimen natural remanent magnetization intensities in the site, in units of mA/m; *N*, total number of specimens demagnetized in site; *e* + *g*, number of endpoint (*e*) and great-circle (*g*) specimens useable for site mean ChRM calculation. Site mean ChRM directions are given by: Dec., declination; Inc., inclination; α_{95}, radius of cone of 95% confidence; *k*, precision parameter estimate. Sites with *italicized* means are not included in the calculation of the overall collection mean direction—see notes.

[a] Site has too few ChRM-bearing specimens to calculate a site mean, and the site is excluded from further analysis.

[b] Site MR26 is combined with site MR27 to represent the basement gneiss.

ing that multidomain magnetite is likely a significant remanence carrier in most sites. Multidomain magnetite can acquire remanence at relatively low temperatures (~200 °C), but laboratory unblocking temperatures of ~450 °C may be required to remove the same remanence (Dunlop, thus significantly overlapping the useful remanence information carried by pseudosingle and single domain magnetite.

Following these observations of the pilot specimens' demagnetization behaviour, other specimens were "AF-cleaned" in three to five demagnetization steps up to 20 mT, and then stepwise thermally demagnetized to reveal their characteristic remanent magnetizations (Fig. 5). Some specimens from the Chatham-Grenville stock underwent low-temperature demagnetization (LTD; Dunlop and Argyle, 1991; Yu et al., 2003) via cycling the specimens through the Verwey transition of magnetite using a liquid N_2 bath. These specimens typically required at least three liquid N_2 cycles to remove most of the viscous remanence, and after three to five LTD cycles were then thermally demagnetized.

ChRM directions were determined using specimens that exhibited stable remanence at unblocking temperatures of >450 °C and coercivities of >20 mT. ChRM directions were identified using the least-squares line-fitting method over four or more consecutive AF or thermal steps (Kirschvink, 1980), where unbiased demagnetization trends could be clearly defined, and by great circle analysis (McFadden and McElhinny, 1988)

Table 2b
Site mean remanence data for the 533 Ma Chatham-Grenville stock, Quebec (45.6°N, 74.50°W)

Site	Rocktype	Intr. (unit)	NRM (mA/m)	*N*	*e* + *g*	ChRM direction			
						Dec. (°)	Inc. (°)	a_{95} (°)	*k*
CG01	Gneiss	–	300	8	–				
CG02	Aplite dyke	1	80	8	–				
CG03	hb syenite	1	3190	3	–				
CG04[a]	Fine hb syenite	1	578	4	2 + 0	*74.6*	*54.8*	–	–
CG05	Grenville dyke	–	2700	8	–				
CG06	Grenville dyke	–	>800	7	5 + 1	95.6	31.6	3.2	460
CG07	Aplite dyke	1	360	12	6 + 2	87.4	23.5	8.0	50
CG08	hb syenite	1	277	9	5 + 1	83.2	22.3	7.6	82
CG09	Fine red dyke	1	500	8	–				
CG10	hb syenite	1	467	9	7 + 0	94.6	18.7	6.8	80
CG11	Grenville dyke	–	656	12	2 + 4	87.9	34.6	16.9	20
CG12	Fine hb syenite	1	272	12	–				
CG13	hb syenite	1	1170	9	–				
CG14	hb syenite	2	1000	7	–				
CG15	Fine hb syenite	2	462	8	–				
CG16	hb syenite	2	314	9	–				
CG17	hb syenite	2	497	8	–				
CG18	Pink syenite	2	279	12	–				
CG19	Fine breccia	2.5	227	13	8 + 1	57.9	35.8	11.7	20
CG20	Fine breccia	2.5	72	10	–				
CG21	Granite	4	3	11	–				
CG22[b]	Granite	4	2	9	3 + 0	*137.0*	*22.9*	*12.7*	*95*
CG23	Granite	4	1	8	–				
CG24	Pink syenite	2	174	8	–				
CG25	Pink syenite	2	>3000	2	–				
CG26	Pink syenite	2	2400	2	–				
CG27	hb syenite	1	42	3	–				
CG28	Grenville dyke	–	>8000	4	–				
CG29	Mafic dyke	1?	326	12	5 + 0	87.6	41.1	11.6	44
CG30	Gneiss	–	30	5	–				
CG31	hb granite	3	18	5	–				
6 Chatham-Grenville sites, excluding CG19						89.7	28.7	7.9	72
7 Chatham-Grenville sites						**85.5**	**30.2**	**10.2**	**35**

Rocktype: hb, hornblende; Intr. unit, location of site within Chatham-Grenville intrusive units 1–4; NRM, mean of specimen natural remanent magnetization intensities in the site, in units of mA/m; *N*, total number of specimens demagnetized in site; *e* + *g*, number of endpoint (*e*) and great-circle (*g*) specimens useable for site mean ChRM calculation. Site mean ChRM directions are given by: Dec., declination; Inc., inclination; α_{95}., radius of cone of 95% confidence; *k*, precision parameter estimate. Sites with *italicized* means are not included in the calculation of the overall collection mean direction—see notes.

[a] Site has too few ChRM-bearing specimens to calculate a site mean, and the site is excluded from further analysis.

[b] Site CG22 appears to have an unusual ChRM as discussed in the text, and is not used in calculation of the combined Chatham-Grenville mean ChRM direction.

where some portion of the viscous remanence overlapped the ChRM.

4.2. Paleomagnetic results

Fig. 5 sets out examples of characteristic demagnetization behaviour for specimens from the Mont Rigaud and Chatham-Grenville stocks and host rocks. An easily-removed northerly and down remanence direction was apparent in most specimens (Fig. 5). It has an overall mean direction of $D = 4.9°$, $I = 69.8°$, which is similar to the expected dipole present-day field direction of 0°, 70° (the observed present-day field direction during sampling was 345°, 72°). We conclude that the northerly and downwards-directed low coercivity and unblocking temperature remanence is of recent origin.

For many specimens AF demagnetization to 20 mT tended to remove nearly all remanence. These specimens exhibited the northerly, downwards VRM only

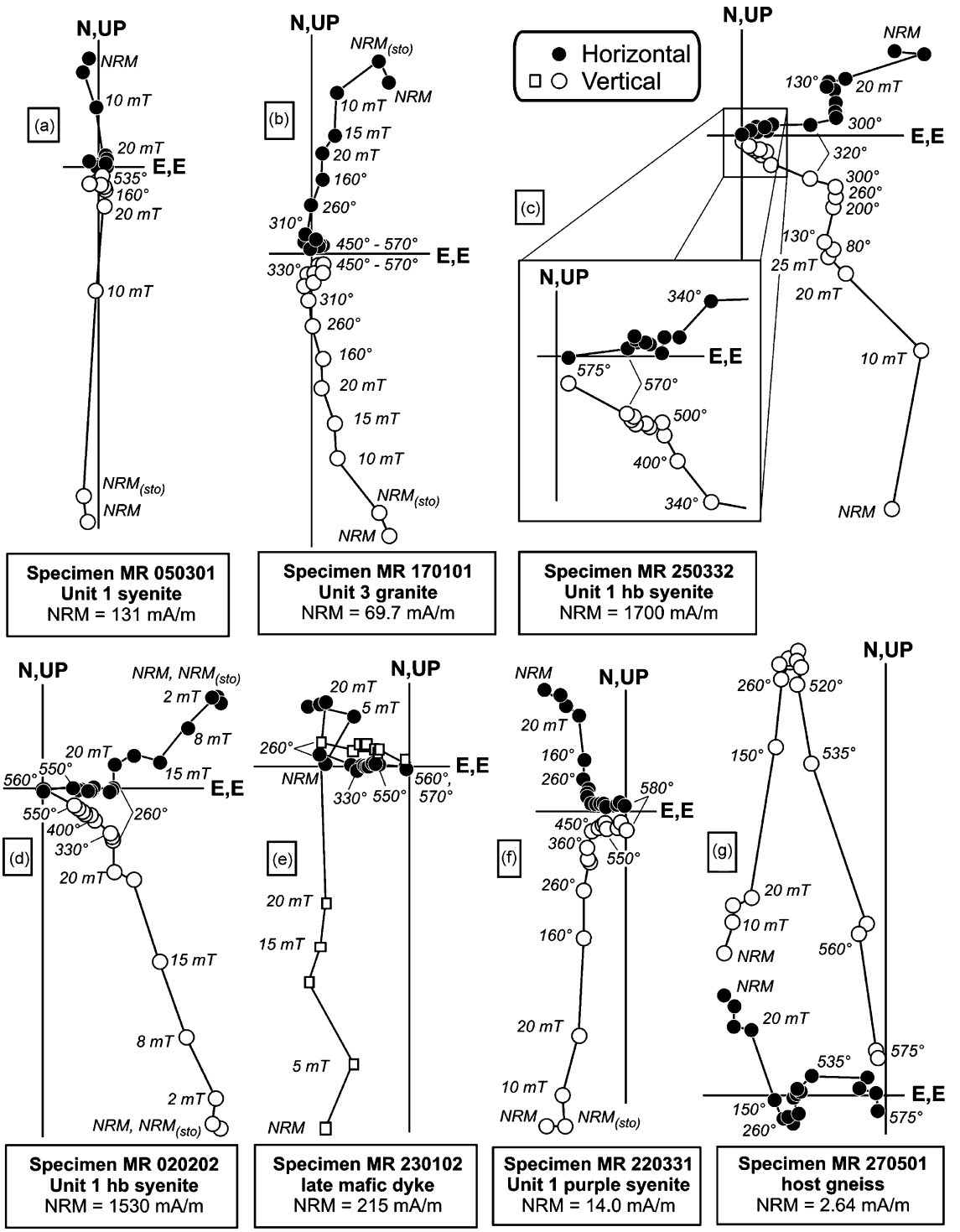


Fig. 5. Orthogonal demagnetization vector plots (Zijderveld, 1967) for representative specimens from the Mont Rigaud (a–g) and Chatham-Grenville (h–m) collections. See text for discussion. Filled circles mark the stepwise demagnetization vectors projected into the horizontal north-south, west-east plane. Open circles or squares mark the same measured demagnetization steps projected into the vertical, west-east plane. LTD# refers to low temperature demagnetization steps (Dunlop and Argyle, 1991) in three to five liquid nitrogen cooling cycles; NRM is the natural remanent magnetization, whose intensity is given for each specimen; $NRM_{(sto)}$ refers to a new NRM measurement made after 1 year or storage in the low-field room to assess the stability of the original NRM.

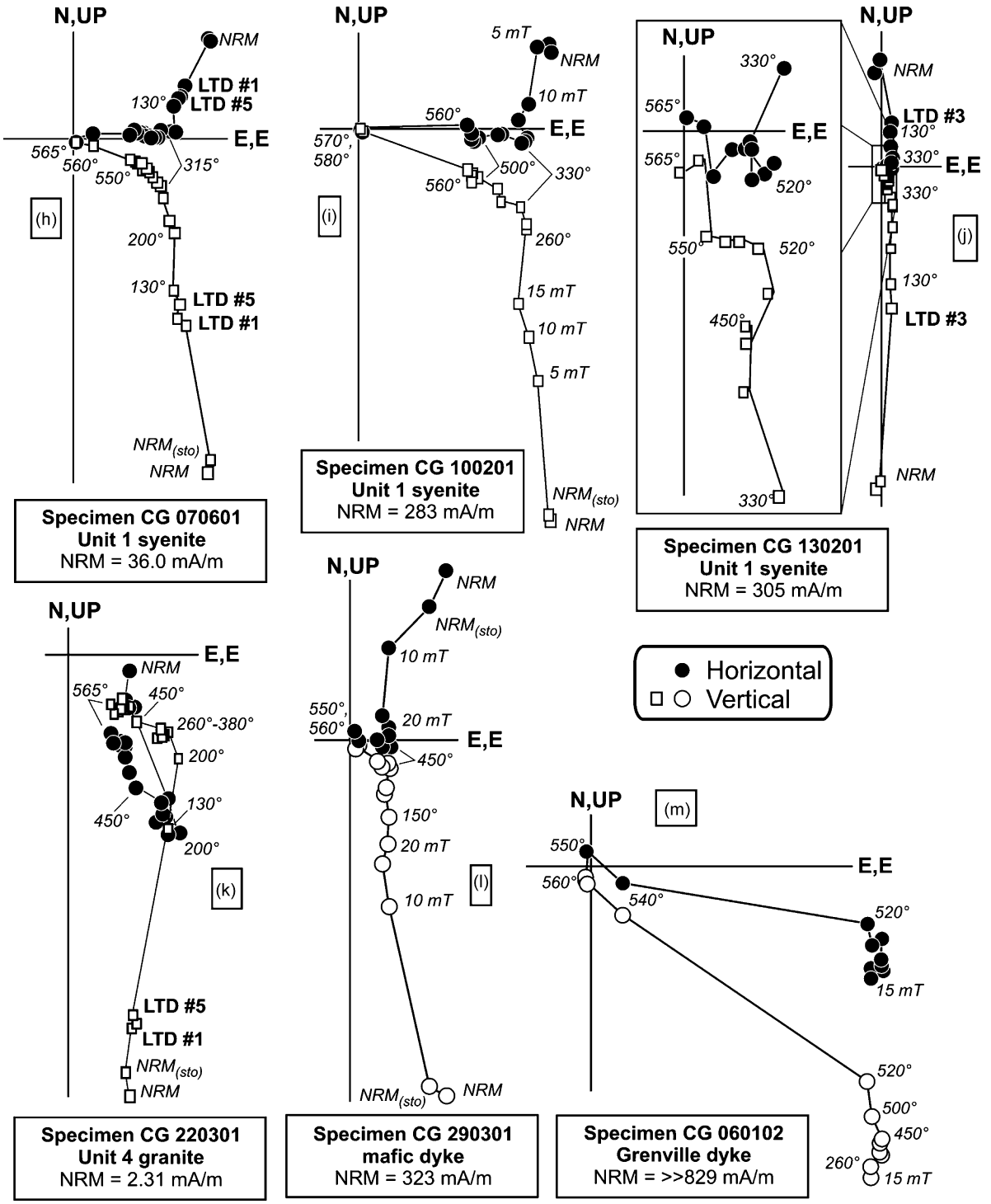


Fig. 5 (*Continued*).

and did not yield any useful ancient remanence, despite further AF or thermal demagnetization (Fig. 5a). Some low-intensity NRM sites carried a northerly, down remanence which survived AF cleaning, but was removed during distributed unblocking to 330 °C, suggestive of maghemite as the remanence carrier (Fig. 5b). More than half of the sampled sites exhibited such soft or low unblocking temperature specimen rema-

nence behaviour, precluding their use in further analysis (Tables 2a and 2b).

In specimens from the more mafic unit 1 syenites of each intrusion, AF demagnetization to 15 or 20 mT, followed by thermal demagnetization to 260° tended to remove the northerly, steep down VRM, revealing a shallowly-directed easterly or westerly ChRM "A" direction that with increasing demagnetizing steps decayed progressively towards the origin of orthogonal vector plots (Fig. 5c–f, h and i). ChRM A is typically defined over unblocking temperatures >450 °C, and characteristically shows a sharp intensity drop in the 550–575 °C range, consistent with PSD magnetite being the remanence carrier. This shallow east or west, high unblocking temperature ChRM A is present in specimens from 17 sites, and has a well-defined mean direction (method of Fisher, 1953; McFadden and McElhinny, 1988) in 15 of those sites (Tables 2a and 2b).

For the Mont Rigaud stock, ChRM A is defined in eight sites out of the 27 sites sampled (Table 2a). Site MR10, which bears ChRM A, is from the unit 2 syenite chill margin on intrusive unit 1; all other ChRM A sites are from unit 1 (Fig. 2a). ChRM A sites MR23 (mafic dyke) and MR22 (purple feldspar syenite) in the Mont Rigaud stock exhibited an opposite polarity westerly, shallow upwards ChRM. Dyke site MR23 was collected as a prospective baked contact test, considering the late stage co-magmatic dyke (Greig, 1968) to potentially have captured a reversal of the Earth's geomagnetic field at the time of its intrusion. An antipodal ChRM A was indeed found in the dyke (Fig. 5e), but the host syenite at site MR23 did not record a stable ancient remanence to complete the contact test. Site MR22, which is from an unusual purple feldspar porphyry syenite, carried a westerly ChRM A which notably does not decay directly towards the origin (Fig. 5f), but instead leaves a steep northerly, down component at unblocking temperatures >580 °C. This leftover component is of the present Earth's field direction and is likely carried by hematite, a good candidate for the coercively hard magnetic mineral in the MR22 samples described in the rock magnetic experiments.

A comparison of the the two westerly ChRM A site mean directions with the six easterly ChRM A site mean directions shows that the unit mean ChRM directions from the two groups are indistinguishable at 95% confidence (test of McFadden and Lowes, 1981). We have inverted the two westerly ChRM A site means through the origin to join the six easterly ChRM A site mean directions, and report a combined unit mean ChRM A direction for the Mont Rigaud stock of $D = 92.5°$, $I = 30.0°$; $\alpha_{95} = 8.8°$, $k = 40.5$; $N = 8$ sites (Fig. 6a).

Specimens from the Chatham-Grenville stock exhibit demagnetization behaviour that is similar to those of the Mont Rigaud stock. A steep down, northerly VRM often dominates (Fig. 5j), but an easterly and shallow ChRM A mean direction was found in specimens from seven of 31 sites (Table 2b), from the more mafic unit 1 syenite phase of the intrusion (Fig. 5h and i), but also from three adjacent host rock mafic dykes (Fig. 5l and m). An eighth site, CG04, also has specimens carrying the ChRM A direction but there are too few to calculate a reliable site mean. In the unit 4 granite site CG22, a hematite-borne remanence was found, directed shallowly down towards the southeast (Figs. 5k and 6b). This remanence direction may represent an overprint of Cambro-Ordovician age, based on the direction's similarity with the expected Late Cambrian to Early Ordovician direction for the Quebec region of Laurentia.

Seven ChRM A sites are defined by a sufficient number of specimens to calculate site mean directions, giving a unit mean ChRM A direction for the Chatham-Grenville stock of $D = 85.5°$, $I = 30.2°$; $\alpha_{95} = 10.2°$, $k = 35.0$; $N = 7$ sites (Fig. 6b). The mean direction for site CG19 is included in the overall mean, although it is somewhat discordant from the other site means. CG19 is sampled from a fine matrix syenitic breccia at the contact between syenite units 1 and 2 which bears clasts of syenite and of the Morin gneiss; the fine breccia is interpreted to have been emplaced rapidly in a subvolcanic vent (Higgins, 1985). The CG19 fine breccia site then may best be viewed as being similar to a single volcanic flow which has not averaged paleosecular variation as well as the slower-cooled intrusive sites.

The unit mean ChRM directions from the two intrusions are indistinguishable at 95% confidence (McFadden and Lowes, 1981), and the two intrusions share a similar geochemistry and are of the same age. We therefore report the Mont Rigaud and Chatham-Grenville paleomagnetic results as one combined result, based on 15 sites in total, with a mean direction of $D = 89.3°$, $I = 30.1°$; $\alpha_{95} = 6.2°$, $k = 38.8$; $N = 15$ sites (Fig. 6c).

4.3. Tests for primary remanence

Several contact tests were sampled for the two intrusions, from an internal, crosscutting mafic dyke (Mont Rigaud site MR23), and from the baked zone of host gneiss and two 590 Ma Grenville dykes adjacent to unit 1 of the Chatham-Grenville stock. In all cases the baked contact tests are inconclusive, due to the lack of a stable, non-VRM dominated remanence in the host rocks. Sites 5, 6 and 11 from the Chatham-Grenville collection

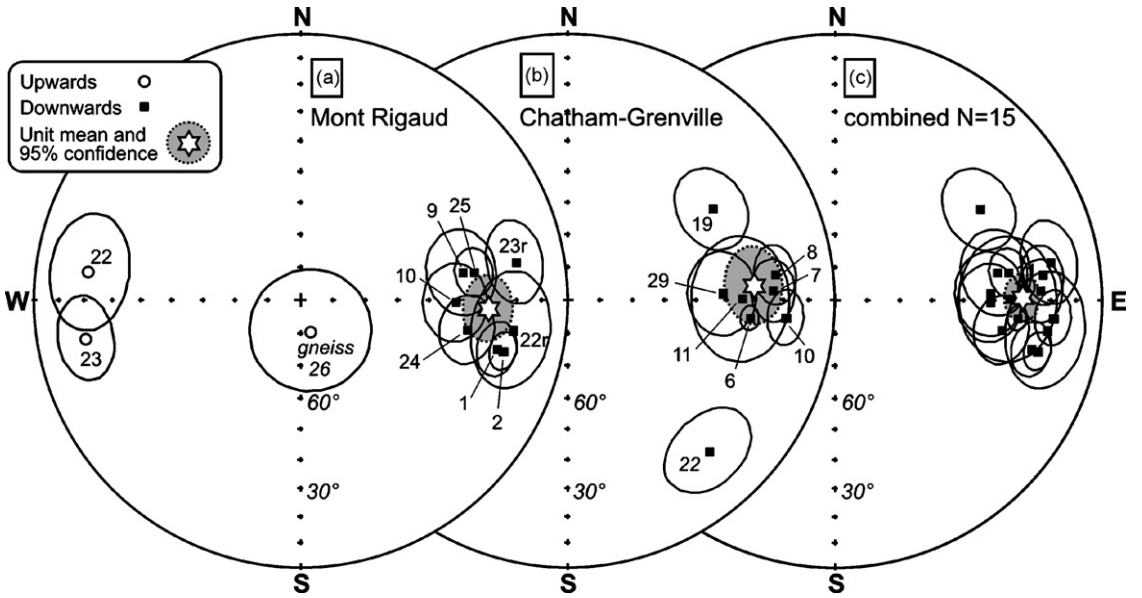


Fig. 6. Equal-area stereographic projections for the site mean directions and their 95% confidence ellipses (Fisher, 1953; McFadden and McElhinny, 1988) for: (a) the Mont Rigaud stock; (b) the Chatham-Grenville stock; (c) combined ChRM A site means from both stocks ($N = 15$ sites).

are sampled from two Grenville dykes within 2–20 m of their contact with the intrusion's unit 1 syenite. Both ca. 590 Ma dykes retain an easterly, down ChRM similar to that of the adjacent intrusion (Fig. 5m), and so are likely within the baked zone of the ca. 533 Ma Chatham-Grenville stock. The more distal Grenville dyke site CG28 retains an exceptionally intense magnetization that yields no consistent direction from sample to sample.

A comparison may also be made between the ChRM exhibited by the Mont Rigaud intrusion with that carried by its gneissic host rock exposed 2 km to the southeast (Fig. 2a). Specimens from Mont Rigaud sites 26 and 27 in the gneiss retain a stable magnetite-borne remanence that is, unlike ChRM A, directed steeply upwards (Figs. 5g and 6a), consistent with the ca. 1100 Ma direction expected for Laurentia in the Rigaud region ($D = 120$, $I = -62°$; Buchan et al., 2000), and with 1100–1060 Ma ^{40}Ar–^{39}Ar cooling ages obtained from metamorphic hornblendes within the Frontenac-Adirondack Belt (Easton, 2000). The presence of the stable, non-ChRM A direction in the host rocks suggests that crystalline rocks in the Mont Rigaud area have not suffered a regional chemical or thermal overprint.

In general, the shallow easterly (westerly) ChRM A directions found in the Mont Rigaud and Chatham-Grenville stocks are unlike any that would have been experienced by this region of Laurentia in post-Cambrian times (Van der Voo, 1993). It is therefore unlikely that ChRM A is an overprint magnetization. Based on its directional dissimilarity with the subsequent Phanerozoic record for Laurentia, the dual-polarity ChRM A found in the Mont Rigaud and Chatham-Grenville stocks is likely to be coeval and primary, representing cratonic Laurentia at ca. 532 Ma with a (north) paleopole position of 11.9°S 184.6°E; $dp = 3.8°$ $dm = 6.9°$. This paleomagnetic result places the west Quebec sampling region at a paleolatitude of 16.2 ± 3.8°S in mid-Early Cambrian time (Fig. 7).

5. Discussion

Paleogeographic relations of the major continental masses and their apparent polar wander (APW) in Late Precambrian to Cambrian time are still poorly defined, in large part due to the lack of well-constrained paleomagnetic results. Here we will discuss the import of the new Early Cambrian paleomagnetic result presented above for Laurentia's APW during the Late Neoproterozoic–Cambrian period and then assess its possible paleogeographic relations with other circum-Iapetus Ocean continents.

5.1. Apparent polar wander of Laurentia in the Ediacaran–Cambrian periods

Laurentia's Late Precambrian and Cambrian APW has been the subject of much recent debate (Pisarevsky et al., 2001; Meert and Torsvik, 2003; Hodych et al., 2004; Murphy et al., 2004; Cawood and Pisarevsky, 2006) con-

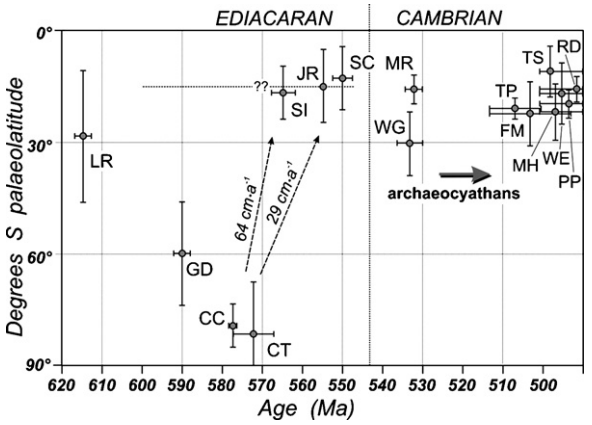


Fig. 7. Paleolatitude vs. age plot for a central location on Laurentia's Iapetan margin (Montreal; 45.5°N, 74.1°W) from 600 to 490 Ma. Unit abbreviations, source paleopoles and their references are given in Table 3. Rapid apparent polar wander (APW) is apparent as the large latitudinal motion from ca. 570 to ca. 560 Ma, and is indicated by the dashed arrows which depict calculated APW rates for two cases: a very rapid 64 cm a^{-1} for the Callander Complex (CC; 577 Ma) to Sept Iles (SI; 564 Ma), assuming the SI magnetization age to be ca. 565 Ma; a rapid 29 cm a^{-1} for the Callander Complex to Skinner Cove (SC; 550 Ma). The arrow labelled 'archaeocyathans' denotes a possible low paleolatidude constraint for Laurentia from the presence of warm-water archaeocyathan reefs on Laurentia's Iapetan margin during Atdabanian time (Williams and Hiscott, 1987).

cerning the evidence for and timing of rapid APW of Laurentia. A compilation of 590–490 Ma paleomagnetic poles for Laurentia is given in Table 3, drawn from the Global Paleomagnetic database v.4.5 (Pisarevsky, 2005), with minor updates as noted. The latitudinal motion of a central location on Laurentia's incipient Iapetan margin (Montreal; 45.5°N, 74.1°W) from 600 to 490 Ma can be gleaned from Fig. 7, and paleopoles from Table 3 are plotted as south poles for Laurentia's APW path in Fig. 8. The most striking feature of this compilation is the large APW and latitudinal motion apparent for Laurentia during Ediacaran time, and the lack of such latitudinal motion in Cambrian time.

Before embarking on a discussion of Laurentian APW it is prudent to discuss the strengths and weaknesses of the paleomagnetic data upon which it is based. We have chosen to tabulate paleopoles which have a quality value (Q, maximally 7; Van der Voo, 1990) of four or greater for rock units which have a demonstrated relationship to Laurentia at the time of their remanence acquisition. We emphasize that the timing of remanence acquisition must also be known to within ±5 Ma to meaningfully address questions of rapid APW. Some exceptions to this criterion are listed in Table 3 for the purpose of discussion, but they are not essential to the first-order description of Laurentia's APW shown in Figs. 7 and 8. Individual paleopoles from Laurentia are assessed below, in order of apparent age grouping.

5.2. Early Ediacaran paleopoles

A paleomagnetic study from the 615 Ma Long Range dykes, Labrador (Murthy et al., 1992) is included in Table 3 but is of limited use due to the ambiguity of its paleomagnetic results, as attested to by the extensive discussion and reinterpretations of the dykes' remanence (Murthy et al., 1992; Meert et al., 1994; McCausland and Hodych, 1998; Hodych et al., 2004; Murphy et al., 2004). The cases for lower latitude (Murthy et al., 1992; Hodych et al., 2004) or high latitude (Meert et al., 1994; McCausland and Hodych, 1998) have been discussed elsewhere (e.g., Murphy et al., 2004; Cawood and Pisarevsky, 2006) to the extent that the existing limited data from the six dykes allow. Pending a more developed paleomagnetic result from the Long Range dykes, we call upon the result as recalculated by Hodych et al. (2004) from five of the six dykes. This recalculation has the advantage of encompassing two polarities but has poor resolution. The Long Range dykes result nevertheless suggests that Laurentia lay at low to moderate paleolatitudes at 615 Ma (Fig. 7).

The first age grouping of poles in Table 3 are from the 590 Ma Grenville Dykes 'B' (Murthy, 1971; Buchan et al., 2004), the 577 Ma Callander Complex (Symons and Chiasson, 1991) and the 572 Ma Catoctin basalts 'A' (Meert et al., 1994). All three studies place Laurentia at moderate to high paleolatitudes during the early- to mid-Ediacaran (Fig. 7). The Grenville Dykes 'B' result (Murthy, 1971), has been studied more extensively by Buchan et al. (2004), who report a preliminary confirmation of the primary nature of the steep directions carried by dykes dated at 590 ± 2 Ma (U–Pb zircon, baddeleyite; Kamo et al., 1995). Shallow 'A' directions carried by other Grenville Dykes (Murthy, 1971) do not have a supporting field test, and so their age of remanence acquisition is as yet unclear (Buchan et al., 2004).

The younger Callander and Catoctin 'A' paleopoles are well determined ($Q = 6$; tight age constraints) and can be used to estimate a ca. 575 Ma pole position for Laurentia. Both results have been criticized as being possible steep overprints (Pisarevsky et al., 2001; Hodych et al., 2004), but the lack of similar expected directions from paleopoles anywhere on the Phanerozoic APWP for North America argue against a younger magnetization age. Sites from both the dual-polarity Callander complex and the Catoctin basalts

Table 3
Ediacaran to Cambrian paleomagnetic results for Laurentia

Rock unit, location	*N*	Pole (south)		$A_{95}{}^{\circ}$ or dp°/dm°	*Q*	Unit age (Ma)	REFNO
		°N	°E				
LR: Long Range dykes, Labrador	5	19.0	355.3	17.4	*5*	615 ± 2[a]	2657[b]
GD: Grenville Dykes 'B', QC	12	60.0	240.0	14.0	*3*	590 ± 2[c]	682
CC: Callander Alkaline Complex, ON	26	46.3	301.4	6.0	*6*	577 ± 1[c]	2508
CT: Catoctin Basalts 'A', VA	6	42.0	296.7	17.5	*6*	572 ± 5[d]	2868
SI: Sept Iles intrusion 'A', QC	10	−20.1	321.3	7.1	*4*	564 ± 4[e]	1925
BF: Buckingham flows, QC (*in situ*)	11	0.0	350.9	7.0	*4*	573 ± 32[f]	1870
JR: Johnnie Fm., NV (unrotated)	∼15	−10.0	342.0	8.0	*4*	ca. 555[g]	1216
SC: Skinner Cove Fm, NFLD	10	−16.4	337.5	9.1	*4*	550 ± 3[h]	3214
WG: Wichita granites, OK	18	−2.0	327.0	8.5	*4*	533 ± 3[i]	523
MR: Mont Rigaud and Chatham-Grenville	15	11.9	4.6	4.8	*5*	532 ± 2	This work
TP: Tapeats sandstone, AZ (unrotated)	6	−4.9	337.7	2.5	*4*	513–501	1044
FM: Florida Mountains auriole, NM	22	6.0	349.0	8.0	*6*	503 ± 10	2375
TS: Taum Sauk limestone, MO	4	−4.0	356.0	7.0	*5*	501–488	1284
MH: Moore Hollow Gp, TX	10	0.6	343.0	6.9	*4*	501–488	2383
WE: Welge ss, Wilberns Fm, TX	9	−10.8	338.2	7.8	*5*	501–488	801
PP: Point Peak ls, Wilberns Fm, TX	7	−6.4	338.8	3.7	*5*	501–488	2376
RD: Royer Dolomite, OK	∼6	−12.6	337.3	3.5	*6*	501–472	2289

N: number of paleomagnetic sites in the given study which define its result, with approximate values (∼) determined by dividing the number of reported contributing samples by six; *Q*: quality value (maximum of 7; Van der Voo, 1990); Unit age is given as a radiometric age with ±error or as a stratographic age, with references numbered and given below for ages not appearing in the paleomagnetic reference; REFNO is the paleomagnetic result reference in the Global Paleomagnetic Database (version 4.6; Pisarevsky, 2005).

[a] Kamo et al. (1989) and Kamo and Gower (1994).
[b] Long Range dykes as recalculated for five dykes by Hodych et al. (2004). Age references.
[c] Kamo et al. (1995).
[d] Aleinikoff et al. (1995).
[e] Higgins and van Breemen (1998).
[f] Lafleur and Hogarth (1981).
[g] Meert et al. (1994), but see Hodych et al. (2004) and discussion in text.
[h] Cawood et al. (2001).
[i] Wright et al. (1996).

carry steeply directed remanence that pass baked contact tests and are retained to high coercivities and high unblocking temperatures. The steep remanence is an ancient and likely primary remanence that must be explained.

A dual-polarity shallow remanence borne by hematite is also found in the Catoctin volcanics and is termed the Catoctin 'B' remanence (Meert et al., 1994). The paleopole derived from Catoctin 'B' has been used to argue for a low-latitude ca. 570 Ma position for Laurentia (Pisarevsky et al., 2001; Hodych et al., 2004; Murphy et al., 2004), taking the prefolding Catoctin 'B' to be primary. The Catoctin 'B' paleopole is similar to Cambrian poles for Laurentia, however, and the Catoctin 'B' remanence is constrained only to be older than the Cambro-Ordovician age of folding; it might be a Cambrian prefolding overprint (Meert et al., 1994). The age of Catoctin 'B' is thus poorly constrained and its paleopole is not helpful in the following discussion.

5.3. Late Ediacaran paleopoles

Four poles are tabulated for later Ediacaran time (Table 3), although none of them can provide an unambiguous, well-resolved pole position for Laurentia. They collectively suggest, however, that Laurentia had arrived at lower latitudes during Ediacaran time prior to the Precambrian–Cambrian boundary (c.f. McCausland and Hodych, 1998). The best constraint is offered by the 550 ± 3 Ma (U-Pb zircon) Skinner Cove Formation, western Newfoundland (Fig. 1), which is allocthonous upon Laurentia's Iapetan margin, but was never far removed from Laurentia (Cawood et al., 2001; Hodych et al., 2004). Given that the Ediacaran APW for Laurentia is manifested as a largely latitudinal motion, the strictly paleolatitudinal constraint from the Skinner Cove volcanics is useful for constraining Laurentia's APW. The Skinner Cove result provides a well-resolved paleolatitudinal constraint which places the Montreal region at $12.5 \pm 9.1^{\circ}$S at 550 Ma (Figs. 7 and 8).

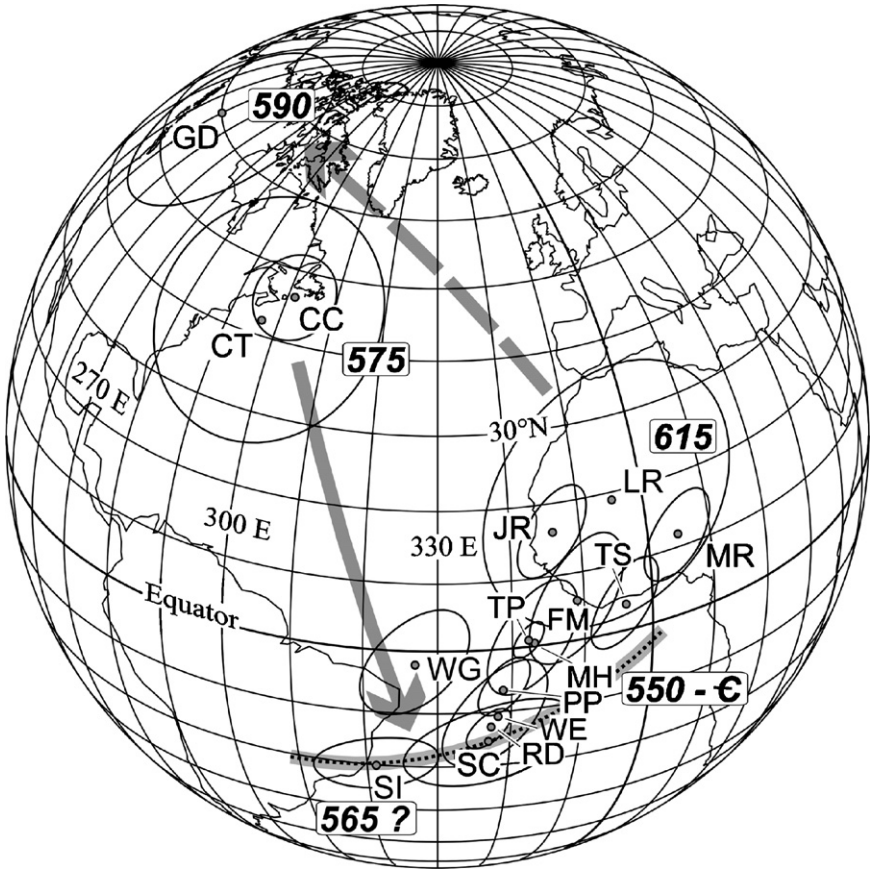


Fig. 8. Ediacaran to Cambrian paleopoles for Laurentia, plotted as south poles with respect to present-day North America. Unit abbreviations and references are given in Table 3. The curved, dashed swath through the Skinner Cove (SC) paleopole indicates the possible range of ca. 550 Ma SC pole positions, given that it comes from an allochthonous unit in Western Newfoundland that is of Laurentian origin (Cawood et al., 2001; Hodych et al., 2004), but may have experienced local net vertical axis rotation. The large grey arrow denotes rapid APW from the ca. 575 Ma poles to the ca. 564 Ma Sept Iles (SI) result, or the ca. 550 Ma Skinner Cove (SC) result, as discussed in the text and shown in Fig. 7.

The other three paleomagnetic results in the later Ediacaran group are part of cratonic Laurentia, but suffer from other uncertainties. The Buckingham flows (Dankers and Lapointe, 1981) have an imprecise age of 573 ± 32 Ma (whole rock K–Ar; Lafleur and Hogarth, 1981) which limits their usefulness in a discussion of Laurentian APW. The characteristic remanence directions carried by the monoclinally southwest-dipping flows are shallow (either *in situ* or tilt corrected) and are similar to those found in the present study of the nearby Mont Rigaud and Chatham-Grenville stocks, raising the possibility that the poorly-dated alkalic Buckingham trachyandesite flows may be a coeval extrusive equivalent of the syenitic Early Cambrian stocks to the east.

Dual-polarity shallow directions are reported from the carbonate Rainstorm member of the Johnnie Formation, Nevada (Van Alstine and Gillett, 1979), but the age of the unit is poorly constrained. Meert et al. (1994) estimated an age of ca. 555 Ma, based on assumed deposition of the overlying Sterling Formation quartzite over 10 million years prior to the Precambrian–Cambrian boundary. An age of 570 ± 10 Ma was proposed by Hodych et al. (2004) based on matching the observed negative $\delta^{13}C$ excursion in the underlying Johnnie oolite unit with the younger of two Varangerian age negative excursions in a standard chemostratigraphic curve (Hoffman and Schrag, 2002). However, Corsetti and Kaufman (2003) document the presence of multiple negative $\delta^{13}C$ excursions in the upper Neoproterozoic of the Death Valley section, including the Johnnie Formation, underscoring the difficulty of matching isotope excursions to idealized chemostratigraphic curves. Indeed, a prominent $\delta^{13}C$ excursion at the top of the Doushantuo Formation, China has an unanticipated age of 555 ± 6 Ma (Zhang et al., 2005), and could well be the equivalent of the carbon isotope excursion found in the upper Johnnie Formation. Until the worldwide Late Neoproterozoic carbon and oxygen isotope record is sufficiently defined, the

matching of isotope excursion records for chronostratigraphy will remain ambiguous (Halverson et al., 2005). In the absence of better age constraints, the Johnnie Formation is given an age of ca. 555 Ma in Table 3 following Meert et al. (1994), but it could be significantly older. Similar age uncertainties trouble the potentially useful paleomagnetic results from other Laurentian Late Precambrian rift-fill successions, such as those from the Double Mer Formation (Murthy et al., 1992), the Bradore Formation (Rao and Deutsch, 1976) and the Blueflower Formation (Park, 1995).

The 565 Ma Sept Iles intrusion paleomagnetic result (Tanczyk et al., 1987) has a shallow, 'A' magnetization present in the intrusion and a steep, dual-polarity 'B' magnetization present in four late mafic dykes as well as in several sites in the intrusion. Tanczyk et al. (1987) interpreted the high coercivity and unblocking temperature 'A' remanence to be primary, and 'B' to be an overprint that notably does not coexist with 'A'. The paleopole derived from the steep 'B' remanence is similar to those of the ca. 575 Ma Callander and Catoctin results, and was interpreted by Symons and Chiasson (1991) to be the primary remanence in the intrusion, with 'A' being a Late Cambrian overprint. In the subsequent literature (e.g., Meert et al., 1994; McCausland and Hodych, 1998; Pisarevsky et al., 2001) either the 'A' or the 'B' paleopoles have been assigned to the intrusive age of the complex (565 ± 4 Ma U–Pb zircon; Higgins and van Breemen, 1998). The 'B' remanence would appear to be younger than 'A,' based on a baked contact test (Tanczyk et al., 1987), but the test is compromised by the strikingly different demagnetization behaviours of the 'B' and 'A' remanences. The 'B' remanence as documented and described by Tanczyk et al. (1987) is much more intense than 'A' and during demagnetization is easily removed by 10–20 mT or exhibits distributed unblocking to ∼500 °C. These features are the hallmarks of remanence carried by multidomain magnetite (Dunlop and so the steep 'B' magnetizations are unlikely to be an ancient remanence despite their dual-polarity. Further study of the Sept Iles intrusion is required to clarify the meaning of the paleomagnetic result, especially to test for a primary thermoremanent nature of the well-defined 'A' magnetization. In Table 3, the Sept Iles 'A' paleopole is assigned the intrusive 565 Ma age, but the 'A' remanence could possibly be a product of younger, Cambro-Ordovician cooling.

5.4. Early Cambrian paleopoles

Two paleopoles are listed in Table 3 for Early Cambrian time, that of the 532 Ma Mont Rigaud (plus Chatham-Grenville) stocks of the present study, and another intrusive result from the 533 Ma Wichita granite, Oklahoma (Spall, 1968; Vincenz et al., 1975). The Mont Rigaud result is deemed to have a Q value of 6, satisfying all quality criteria except for a clearly positive test for primary remanence (Van der Voo, 1990). The Wichita granite and associated Carlton rhyolite have a crystallization age of 533 ± 3 Ma (U–Pb zircon; Wright et al., 1996) and are emplaced with angular unconformity over the Glen Mountains anorthositic layered complex (Hogan et al., 2000). Paleomagnetic study of the poorly dated Glen Mountains complex reveals a stable remanence which is possibly overprinted at one site near contact with the Wichita granite by the characteristic direction of the granite, thus suggesting that Wichita granite may retain a primary remanence (Roggenthen et al., 1981). The paleopole position for the Wichita granite is discordant from that of Mont Rigaud or other Cambrian results, however, and merits further study.

5.5. Mid to Late Cambrian paleopoles

Seven results define the Middle and Late Cambrian APW path for Laurentia. The oldest is from the Middle Cambrian Tapeats sandstone, Arizona (Elston and Bressler, 1977), which, after correction for 7.4° of post-Cretaceous clockwise rotation of the Colorado Plateau (Molina-Garza et al., 1996) provides a well-constrained paleopole not far removed from that of the metaigneous contact auriole of the 503 ± 10 Ma (U–Pb zircon) Florida Mountains intrusion, New Mexico (Geissman et al., 1991). Three well-defined results from the Late Cambrian Moore Hollow Group, Texas (Van der Voo et al., 1976; Loucks and Elmore, 1986; Farr and Gose, 1991) are similar. For clarity we retain these three paleomagnetic database poles as separate Table 3 entries, and also note that each study likely provides a distinct reading of the Late Cambrian paleofield based on their differing lithologies and remanence carriers. The Taum Sauk limestone provides an independent well-constrained Late Cambrian pole, although is is based on only four sites (Dunn and Elmore, 1985). Finally, the Late Cambrian to Early Ordovician Royer dolomite retains a magnetite-borne remanence that is well-resolved from a hematite-borne Carboniferous overprint (Nick and Elmore, 1990), providing a likely primary paleopole.

5.6. Laurentia's role in Cambrian true polar wander

A substantial contribution of true polar wander (TPW – the bulk tumbling of the Earth with respect to its

spin axis) to the apparent motions of continents should manifest itself as simultaneous, long APW tracks for all parts of the Earth that are capable of preserving a paleomagnetic record (Deutsch, 1963; Golderich and Toomre, 1969). Kirschvink et al. (1997) proposed that an inertial-interchange true polar wander (IITPW) event took place during the Early Cambrian, based on a comparison of the then – available paleomagnetic results from major continents which they interpreted to show nearly 90° of APW between 530 and 515 Ma. The IITPW event in its simplest form requires ~90° of simultaneous APW track from each continent because the maximum and intermediate moments of inertia become subequal and exchange identities; the Earth then tumbles through 90° about its minimum inertial moment axis, taking the new maximum inertial moment from the equator to the poles (Golderich and Toomre, 1969; Chase, 1985). The proposed IITPW event is testable with well-constrained paleomagnetic poles which define each continent's APW during the Early Cambrian (Meert, 1999).

Evans (1998, 2003) recognized that the small mass redistribution that causes the exchange of subequal intermediate and maximum inertial moments could be reversible, leading to an "ocscillatory TPW," in which the Earth tumbles unstably between these two subequal moments until a new maximum inertial moment is clearly established and aligned with the Earth's rotational poles. In this case APW tracks adopt a zig-zag pattern within the overall TPW-dominated track, a situation that is perforce much more difficult to test, even with well-constrained and dated paleomagnetic poles.

In the Kirschvink et al. (1997) proposal for Cambrian IITPW, Laurentia is postulated to have been at the south pole until the end of the Nemakit-Daldynian stage (at ca. 530 Ma; Gradstein et al., 2004) and then travelled northwards during the IITPW event, arriving at the equator by ca. 513 Ma, the start of Middle Cambrian time. In this study we have found that Laurentia straddled low paleolatitudes in Nemakit-Daldynian time at 532 Ma, similar to its Late Ediacaran 550 Ma latitudinal position and to its Middle and Late Cambrian latitudinal positions (Fig. 7). These paleomagnetic results most simply indicate that Laurentia resided at the equator throughout Cambrian time, and further suggest that Laurentia had at best only moderate APW during this period (Fig. 8), thus refuting IITPW for Cambrian time. An oscillatory TPW event (Evans, 1998, 2003) is still possible for times between Laurentia's 550 Ma, 532 Ma and Middle Cambrian published paleomagnetic results, but is not required or even suggested by them.

5.7. Circum-Iapetus paleogeography at the Precambrian–Cambrian boundary

The Iapetus Ocean that was originally defined by Harland and Gayer (1972) as being the Early Paleozoic oceanic tract which lay between the margins of East Greenland and Norway may no longer have a clear paleogeographic meaning. It is still useful, however, to consider the oceans that opened during the breakup of Rodinia and that closed during the early stages of the assembly of Pangea to be broadly 'Iapetan,' at least until a more clearly defined paleogeography can be determined.

Figs. 9 and 10 show two possible paleogeographies for the Precambrian–Cambrian transition, based on the Laurentian paleomagnetic results compiled here (Table 3) and published results for other major presumed circum-Iapetus continents (Table 4) as summarized for the West Gondwanan elements by Tohver et al. (2006a), for Baltica by Cawood and Pisarevsky (2006), and for Siberia by Kravchinsky et al. (2001) and Meert and Torsvik (2003). In all cases relative longitudes between the continents is unconstrained, thus leading to some unavoidable ambiguity in the interpreted paleogeography. Paleogeographic interpretations based on relative paleolatitudes, however, are much more robust (Table 5).

The essential difference between the paleogeograpies depicted in Figs. 9 and 10 is in the interpretation of the paleomagnetic and geological record of Baltica; Fig. 9 shows Baltica in its inverted, or "upside down" orientation (Hartz and Torsvik, 2002), whereas Fig. 10 has Baltica in a "rightside-up" orientation (Hoffman, 1991; Cawood and Pisarevsky, 2006). The role of Baltica in the paleogeography of the Precambrian–Cambrian transition will be discussed further below.

Figs. 9 and 10 show a wide Iapetus Ocean in Late Ediacaran time (550 Ma) and Early Cambrian time (530 Ma). A comparison of Laurentian and Gondwanan paleopoles for Late Ediacaran time shows that Laurentia was at low paleolatitudes whereas its postulated conjugate "proto-Andean" rift margin of West Gondwana (e.g., Dalziel, 1997) resided at high southerly paleolatitudes during the same time (Figs. 9a and 10a), indicating that the ephemeral supercontinent Pannotia (Laurentia combined with an assembled Gondwana; Powell, 1995; Dalziel, 1997) could not have existed.

For earlier Ediacaran time, the elements of West Gondwana consistently appear to have been separated from Laurentia: West Gondwanan paleopoles for ca. 600–580 Ma (Table 4) are broadly in agreement, although the elements of West Gondwana may not have been fully amalgamated until Cambrian time (Tohver et

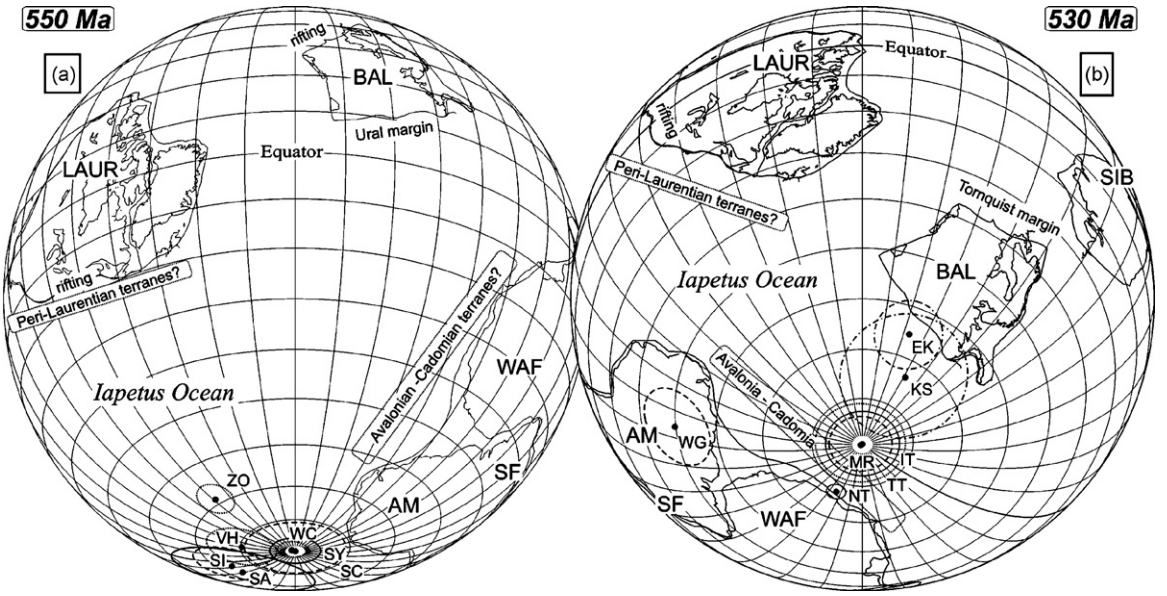


Fig. 9. Circum-Iapetus paleogeography, with Baltica in an inverted orientation (Hartz and Torsvik, 2002), for: (a) Late Ediacaran, 550 Ma; (b) Early Cambrian, 530 Ma. Paleomagnetic pole abbreviations and other information are drawn from Tables 3 and 4. West Gondwana paleopoles are circumscribed by their A_{95} error limits in solid lines, poles for Laurentia in dashed lines, for Baltica in dotted lines, and for Siberia in dash-dot alternating lines. Continents are shown with their present-day coastlines for recognition only; West Gondwana (outlines of Africa and South America) is depicted as completely assembled although it may not have been until Early Cambrian time (Trindade et al., 2006). Continent abbreviations are: AM, Amazonia; BAL, Baltica; LAUR, Laurentia; RP, Rio de la Plata; SF, Sao Francisco; WAF, West Africa. Euler rotations for the reconstructions are given in Table 5.

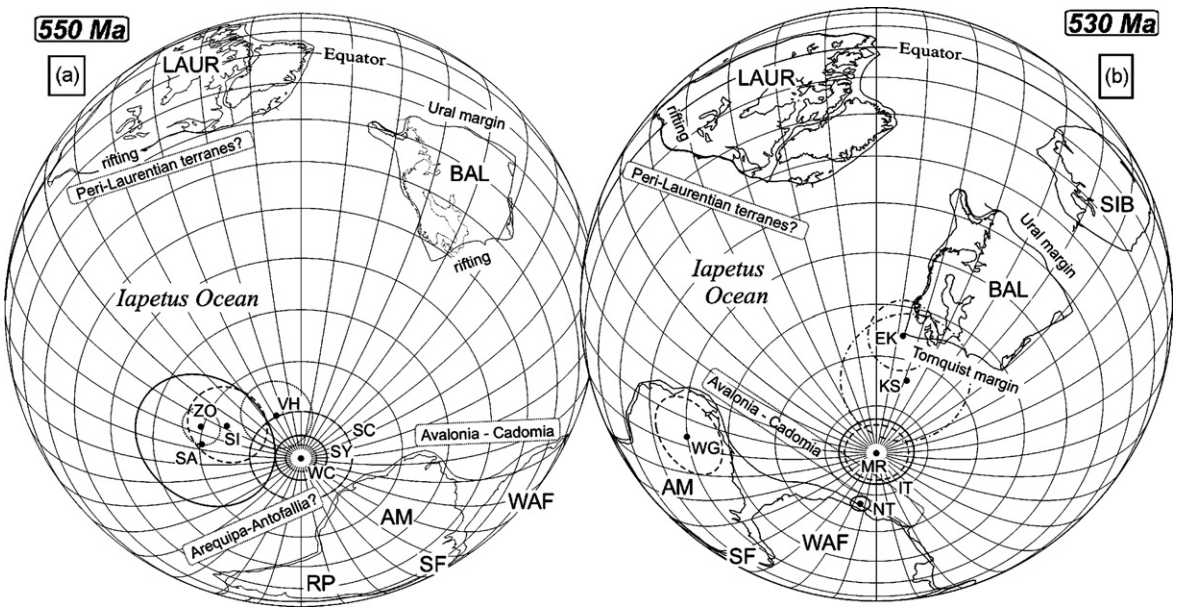


Fig. 10. Circum-Iapetus paleogeography, with Baltica in an "rightside-up" orientation (Hoffman, 1991; Cawood and Pisarevsky, 2006), for: (a) Late Ediacaran, 550 Ma; (b) Early Cambrian, 530 Ma. Paleomagnetic pole abbreviations and other information are drawn from Tables 3 and 4. West Gondwana paleopoles are circumscribed by their A_{95} error limits in solid lines, poles for Laurentia in dashed lines, for Baltica in dotted lines, and for Siberia in dash-dot alternating lines. Continents are shown with their present-day coastlines for recognition only; West Gondwana (outlines of Africa and South America) is depicted as completely assembled although it may not have been until Early Cambrian time (Trindade et al., 2006). Continent abbreviations are given in Fig. 9 and Euler rotations for the reconstructions are given in Table 5.

Table 4
Ediacaran to Cambrian paleomagnetic results for West Gondwana, Baltica and Siberia

Rock unit, location	*N*	Pole (south)		A_{95}° or d*p*°/d*m*°	*Q*	Unit age (Ma)	REFNO
		°N	°E				
West Gondwana (reported in African coordinates,[a] from Tohver et al., 2006a)							
AD: Adma diorite, Mali (*WAF-Pharus*)	13	33.5	343.5	16.5	*3*	613 ± 3	1170
HS: Huqf dolom., siltst., Oman (*ANS*)	12	52.3	254.4	7.3	*6*	∼600	[b]
PG: Puga cap carbonate (*AM*)	13	51.0	249.0	8.0	*4*	∼600	3669
CA: Campo Allegre rhyolites (*RP*)	6	82.0	233.6	11.7	*6*	595 ± 5	1369
DK: Dokhan volcanics, Egypt (*ANS*)	10	43.0	216.2	10.0	*4*	593 ± 13	3101
SA: Sierra de las Animas 2 syenite (*RP*)	6	−46.8	312.0	22.0	*5*	540–560	3513
SY: Sinyai metadolerite, Kenya (*Moz*)	∼8	−28.0	319.0	5.0	*5*	547 ± 4	3106
IB: Itabaiana dykes (*Brazil*)	15	31.4	331.0	10.0	*6*	525 ± 5	[c]
NT: Ntonya Ring, Malawi (*WAF*)	∼5	27.7	344.9	1.8	*4*	522 ± 13	388
Baltica (from Cawood and Pisarevsky, 2006)							
ED: Egersund dykes, Norway	4	22.0	33.0	19.0	*4*	616 ± 3	447, 689
FC: Fen carbonatite complex, Norway	6	56.0	330.0	9.0	*4*	583 ± 15	3447
WC: Winter Coast shale, Russia	∼15	25.0	312.0	3.0	*6*	555 ± 1	3506
VH: Volhynia volcanics, Ukraine	6	34.0	306.0	–	*4*	551 ± 4	[d]
ZO: Zolotitsa red clays, Russia	∼9	28.3	290.0	3.8	*6*	550 ± 5	[e]
TT: Torneträsk Fm, Sweden	6	56.0	116.0	12.0/15.0	*4*	543–520	3425
Siberia (from Kravchinsky et al., 2001; Meert and Torsvik, 2003)							
SH: Shaman red ss (*SW Sib*)	10	32.0	251.1	6.9/13.8	*4*	∼600	3421
MY: Minya sandstone (*SW Sib*)	∼2	33.7	217.2	8.6/14.7	*3*	∼600	3421
CB: Cisbaikalia red ss (*S Sib*)	∼5	−2.7	168.2	5.5/10.1	*5*	∼600	3301
KS: Kessyusa sandstone (*NE Sib*)	∼2	−37.6	165.0	9.3/15.4	*4*	525–530	3164
EK: Erketket red limestone (*NE Sib*)	∼4	−44.8	158.7	5.7/8.8	*4*	520–525	3164
KT: Khorb. Toyonian Erketket (*NE Sib*)	∼6	−53.3	135.0	5.1/7.2	*5*	513–520	3537
KA: Khorb. Amgan Yuryakh (*NE Sib*)	∼9	−45.5	136.6	3.7/5.8	*6*	508–513	3537
KM: Khorb. Mayan Yuryakh (*NE Sib*)	∼16	−42.7	132.5	2.5/4.0	*6*	500–508	3537
YY: Yuryakh Mayan (*NE Sib*)	∼6	−36.4	139.6	3.0/5.2	*6*	500–508	3164
KL: Kulumbe R. limestone (*NW Sib*)	∼21	−41.9	136.2	1.8/2.9	*6*	500–508	[f]

Rock unit locations are given regional modifiers (*in italics*) for West Gondwana and Siberia, recognizing that these continents may not have been fully assembled throughout Ediacaran-Cambrian time (Tohver et al., 2006a; Kravchinsky et al., 2001). Abbreviations are: AM, Amazonia; ANS, Arabian-Nubian shield; Brazil, Braziliano belt; Moz, Mozambique belt; RP, Rio de la Plata; WAF-Pharus, West Africa-Pharusian belt. *N*: number of paleomagnetic sites in the given study which define its result, with approximate values (∼) determined by dividing the number of reported contributing samples by six; *Q*: quality value (maximum of 7; Van der Voo, 1990); Unit age is given as a radiometric age with ±error or as a stratographic age; REFNO is the paleomagnetic result reference in the Global Paleomagnetic Database (version 4.6; Pisarevsky, 2005). Letters in the REFNO column mark results which are not in the database, and are referenced as follows.

[a] South America rotated to Africa (WAF) coordinates by the Euler rotation pole 45.5°N, 327.8°E, 58.2°CCW (Lawver and Scotese, 1987).
[b] Kilner et al. (2005).
[c] Trindade et al. (2006).
[d] Nawrocki et al. (2004).
[e] Iglesia-Llanos et al. (2005).
[f] Pavlov and Gallet (2001).

al., 2006a; Trindade et al., 2006). The ca. 600–580 Ma West Gondwanan poles, including the result from the Puga carbonate (Trindade et al., 2003) place Amazonia at low paleolatitude during the same time that ca. 590–570 Ma paleopoles place Laurentia in the polar region (Table 3). The Puga result directly represents Amazonia, but its similarity to the present-day pole position leaves open the question of its age of remanence acquisition (Tohver et al., 2006a,b). The existing, admittedly sparse paleomagnetic results do suggest that Laurentia was well separated from the proto-Andean margin throughout much of Ediacaran time. The large Iapetus Ocean depicted in Figs. 9 and 10 may have been initiated even before Ediacaran time.

There is nevertheless abundant stratigraphic evidence for a rift-to-drift transition at the Precambrian–Cambrian boundary along the Iapetan margin of Laurentia (Bond et al., 1984; Williams and Hiscott, 1987). It is possible that the conjugate rifted margin to Laurentia consisted of terranes rather than of cratons (Cawood et al., 2001),

Table 5
Euler pole rotations for continents in Figs. 9 and 10

Euler operation	Euler pole (rotation)		
	°N	°E	°CCW
Fig. 9a: 550 Ma			
Laurentia to Africa	40.49	168.92	−46.80
Baltica to Africa	74.45	41.21	−159.25
Africa (West Gondwana) to absolute reference frame, using the Sinyai pole (SY in Table 4)	0	49.00	62.00
Fig. 9b: 530 Ma			
Laurentia to Africa	3.71	337.57	85.31
Baltica to Africa	75.62	292.14	−136.85
Siberia to Africa	13.37	330.15	59.05
Africa (West Gondwana) to absolute reference frame, using the Itabaiana pole (IT in Table 4)	0	240.95	−121.4
Fig. 10a: 550 Ma			
Laurentia to Africa	30.89	156.96	−84.20
Baltica to Africa	2.17	345.17	91.04
Africa (West Gondwana) to absolute reference frame, using the Sinyai pole (SY in Table 4)	0	49.00	62.00
Fig. 10b: 530 Ma			
Laurentia to Africa	0.58	335.68	74.65
Baltica to Africa	18.84	343.35	75.59
Siberia to Africa	13.37	330.15	59.05
Africa (West Gondwana) to absolute reference frame, using the Itabaiana pole (IT in Table 4)	0	240.95	−121.4

with candidate terranes being Oaxaquia (Cawood et al., 2001) and peri-Laurentian Appalachian basement terranes such as Dashwoods (Waldron and van Staal, 2001) and Goochland (Bartholomew and Tollo, 2004). If the ca. 550 Ma rifting that occurred along Laurentia's Iapetan margin does not represent the separation of true cratons from Laurentia, then the timing of the birth of the Iapetus Ocean is an open question.

Assuming that a Grenvillian collisional relationship between Amazonia and eastern Laurentia is correct (Hoffman, 1991; Dalziel, 1997; Tohver et al., 2006a,b), then the opening of the Iapetus Ocean between Laurentia and West Gondwana is only constrained to have happened between ca. 920–750 Ma post-Grenville extension (e.g., Dalziel and Soper, 2001; Streepey et al., 2004) and the onset of ca. 570 Ma voluminous rift magmatism (Cawood et al., 2001). The Iapetus Ocean could date from the earlier "pulse" of magmatism along Laurentia's proto-Iapetan margin at ca. 760–680 Ma (Badger and Sinha, 1988; Aleinikoff et al., 1995; Tollo et al., 2004). The 760–680 Ma magmatism is restricted to the the Laurentian margin along the southern Appalachians (Fig. 1) and consists dominantly of felsic, within-plate A-type plutons that suggest a rift setting, although no marine rift-drift sedimentation is recorded for the 760–680 Ma period in the U.S. Appalachians (Tollo et al., 2004). If the Iapetus Ocean began to open as early as ∼750 Ma, then much of its early history may be recorded in what subsequently became peri-Laurentian and peri-Gondwanan terranes rather than in its presumed bordering cratons.

The position and orientation of Baltica throughout the Ediacaran and Cambrian is controversial, with recent interpretations (e.g., Meert and Lieberman, 2004; Cocks and Torsvik, 2005) showing Baltica in an inverted, or "upside down" orientation (Fig. 9a) in which its Uralian margin is interpreted to have rifted from the East Greenland region of Laurentia (Hartz and Torsvik, 2002). Baltica is depicted in an "upside down" orientation in Fig. 9a using the ca. 555 Ma paleomagnetic result from the Winter Coast sediments (Popov et al., 2002) in its reported south polarity option. Baltica is shown "rightside up" for Late Ediacaran time in Fig. 10a using the opposite polarity of the Winter Coast result as a south pole (c.f. Cawood and Pisarevsky, 2006). New ca. 550 Ma Baltica results from different regions (Iglesia-Llanos et al., 2005; Nawrocki et al., 2004) agree with the Winter Coast result (Table 4), suggesting that regardless of polarity choice Baltica is well constrained to have resided at low paleolatitude in Late Ediacaran time. Baltica's low latitude position at ca. 550 Ma does not directly place its Late Neoproterozoic rifted Baltoscandian or Tornquist margins adjacent to Laurentia's Iapetan margin in either of the "upside down" or "rightside up" interpretations (Figs. 9a and 10a), although the "rightside up" option is more geologically plausible (Cawood and Pisarevsky, 2006).

In Early Cambrian time, Baltica's position is uncertain. Two options are shown, one with Baltica "upside down" (Fig. 9b) as given by the Nemakit-Daldynian age Torneträsk siltstone result (Torsvik and Rehnstrom, 2001), and the other with Baltica "rightside up" (Fig. 10b), oriented by an interpolated Baltica paleopole at 32°N, 355°E, between Baltica's ca. 550 poles (Table 4) and the Early Paleozoic (Ordovician) mean Baltica pole of Smethurst et al. (1998). The Torneträsk result has one reversed site out of six but no conclusive test for primary remanence, and its remanence direction is similar to that expected for its location in Late Triassic to Early Jurassic time (Van der Voo, 1993). The Baltica "upside down" interpretation requires that Baltica underwent ∼120° of counter-clockwise rotation in the Late Cambrian–Early Ordovician (Cocks and Torsvik, 2005); whereas the Baltica "rightside up" interpretation is at odds with the Torneträsk result. Well-constrained Cambrian paleomagnetic results from Baltica are needed to resolve its Cambrian paleogeographic relations.

Siberia may or may not have bordered the Iapetus Ocean. It is omitted from the 550 Ma Figs. 9a and 10a due to the lack of constraining paleomagnetic results, but it may have been located at low to mid-latitudes in the northern hemisphere near the proto-Cordilleran margin of Laurentia in its possible Mesoproterozoic position (Pisarevsky and Natapov, 2003) or in direct rifting relationship with Laurentia's Cordilleran margin (Sears and Price, 2003). For 530 Ma, Siberia is shown in inverted orientation (Figs. 9b and 10b), with its northern, Taimyr margin facing generally southwards as suggested by paleomagnetic results from the Tommotian Kessyusa Formation sandstones and Atdabanian Erketket Formation limestones of northeastern Siberia (Pisarevsky et al., 1997). Kravchinsky et al. (2001) highlighted the possibility that the Siberian platform may not have been assembled via relative block rotations until the later Cambrian, so the above referenced results may best be taken to indicate that Siberia resided at or near the equator during the Early Cambrian. The similarity of Eastern Laurentia with Baltica and Siberia in biogeographic relationships for Early to Middle Cambrian trilobites (Meert and Lieberman, 2004), and Ediacaran fauna (Waggoner, 1999) suggests that these continents were associated throughout the Precambrian–Cambrian transition.

West Gondwana exhibits a large apparent polar wander of about 60° over 22 myears (∼30 cm a^{-1}) between its ca. 550 Ma paleopoles and its ca. 525 Ma paleopoles (Table 4; Figs. 9 and 10), as noted by Trindade et al. (2006), but it is unclear how much of this APW might be an artefact of comparing paleomagnetic results from West Gondwanan blocks which had not yet amalgamated (Cordani et al., 2003). A wider consideration of West and East Gondwanan paleomagnetic results for the Ediacaran–Cambrian does support the notion of a mostly coherent Gondwana by Early Cambrian time (Meert and Van der Voo, 1996; Meert and Torsvik, 2003), and thus suggests that much of the large 550–525 Ma APW for West Gondwana has tectonic significance for the West Gondwanan Iapetan margins. In general, the large APW for West Gondwana is coincident with the timing of transform motion for Avalonian and other peri-Gondwanan terranes (Murphy et al., 2004).

5.8. Rapid APW motion during the Ediacaran: an Ediacaran loop?

Laurentia experienced rapid paleolatitudinal motion from the south polar region at ca. 575 Ma to the equator by 550 Ma (Figs. 7 and 8). If the Sept Iles A low inclination result (Tanczyk et al., 1987) is primary and of 565 Ma age, then the Laurentia pole-to-equator motion must have occurred between ca. 575 and 565 Ma, requiring an extraordinarily rapid northwards motion of >60 cm a^{-1}, well beyond the maximum plausible rates of plate motion due to drift alone (Meert et al., 1993). A less rapid rate of 29 cm a^{-1} is implied by comparison of the best constrained paleomagnetic results (the ca. 577 Ma Callander Complex and the ca. 550 Ma Skinner Cove volcanics), and could be a product of unusually rapid plate motion via non-plate boundary mechanisms such as cratonic motion from a geoid high to a geoid low (Gurnis and Torsvik, 1994; Meert, 1999).

The solitary ca. 615 Ma Long Range dykes result (Murthy et al., 1992; Hodych et al., 2004) places Laurentia at low latitude in earliest Ediacaran time. The southwards motion of Laurentia between 615 Ma and its south polar position as marked by the 590 Ma Grenville dykes (Murthy, 1971; Buchan et al., 2004) is 38 cm a^{-1} (85° of APW over 25 myears). Laurentia therefore appears to have had rapid APW throughout the Ediacaran period, moving from low latitude to the south polar region and then back to low latitude by the start of the Cambrian (Fig. 7). Laurentia's apparent polar wander path during the Ediacaran can speculatively be shown as describing a large loop, an "Ediacaran loop" of 50–65 myears duration (Fig. 8).

Any major contribution from true polar wander to the rapid Ediacaran motion of Laurentia must also be

expressed as similarly long APW tracks for other cratons during the same period (Meert, 1999; Evans, 2003). Some evidence exists for long Ediacaran APW tracks for elements of West Gondwana (Sánchez-Bettucci and Rapalini, 2002; Tohver et al., 2006a), Baltica (Popov et al., 2002; Cawood and Pisarevsky, 2006) and Siberia (Kravchinsky et al., 2001), but there are major ambiguities arising from the choice of polarity and especially the poor control on the age of remanence for many of the paleomagnetic results. We note, however, that large (although not necessarily rapid) APW is suggested by the Ediacaran through Cambrian paleopoles from Baltica, Siberia and the West Gondwanan elements regardless of the polarity choice and the exact age of particular results (Tables 3 and 4).

The large Ediacaran APW for Laurentia, elements of West Gondwana, Baltica and Siberia can be interpreted as having recorded Ediacaran TPW (McCausland and Van der Voo, 2004), giving an Ediacaran circum-Iapetus paleogeography that is similar to that depicted for the Precambrian–Cambrian transition in Fig. 9. At this juncture, however, the paleomagnetic record in the major continents is not yet well enough constrained, especially in age, to more rigorously assess the possibility of major Ediacaran TPW.

Acknowledgements

Rachel Jones, Darren Smith, Roberta Flemming and Burt McKinley gave able and friendly field assistance, and Tanya Shavalia, J.P. Brandenburg, Dana Dougherty and Ping Zhu helped with sample preparation and measurements. Paleomagnetic data interpretation was performed using Super-IAPD (Torsvik, Briden and Smethurst) and PMGSC (Enkin) software. Paleogeographic figures have been plotted with the aid of GMAP 2003 (Torsvik) software. This work has benefitted from discussions with Eric Tohver, Ken Buchan, Duncan Keppie and David Symons, and from the comments of two anonymous reviewers. We also thank Maria Cioppa and David Symons for use of facilities at the University of Windsor paleomagnetics laboratory for a portion of this study. Rock magnetic measurements were made during a visiting Fellowship by PJAM at the Institute for Rock Magnetism, University of Minnesota. PJAM also gratefully acknowledges support from a Scott Turner Postdoctoral Fellowship at the Department of Geological Sciences, University of Michigan, as well as from the Division of Earth Sciences and the Office of International Science and Engineering's Eastern and Central Europe Program of the U.S. National Science Foundation, grant EAR 0335882.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.precamres.2007.03.004.

References

Aleinikoff, J.N., et al., 1995. U–Pb ages of metarhyolites of the Catoctin and Mount Rogers formations, central and southern Appalachians: evidence for two pulses of Iapetan rifting. Am. J. Sci. 295, 428–454.

Badger, R.L., Sinha, A.K., 1988. Age and Sr isotopic signature of the Catoctin volcanic province: implications for subcrustal mantle evolution. Geology 16, 692–695.

Bartholomew, M.J., Tollo, R.P., 2004. Northern ancestry for the Goochland terrane as a displaced fragment of Laurentia. Geology 32 (8), 669–672.

Bingen, B., Demaiffe, D., van Breemen, O., 1998. The 616 Ma old Egersund basaltic dike swarm, S.W. Norway and Late Neoproterozoic opening of the Iapetus Ocean. J. Geol. 106, 565–574.

Bond, G.C., Nickerson, P.A., Kominz, M.A., 1984. Breakup of a supercontinent between 625 and 555 Ma: new evidence and implications for continental histories. Earth Planet. Sci. Lett. 70, 325–345.

Bowring, S.A., Erwin, D.H., 1998. A new look at evolutionary rates in deep time. Geol. Today 8, 1–8.

Buchan, K.L., Ernst, R.E., Kumarapeli, P.S., 2004. Paleomagnetism of the Grenville diabase dyke swarm and implications for the mid Vendian paleolatitude of Laurentia. EOS: Trans. Am. Geophys. Union 85 (17) (Abstract GP24A-02, JA166).

Buchan, K.L., Mertanen, S., Park, R.G., Pesonen, L.J., Elming, S.-A., Abrahamsen, N., Bylund, G., 2000. Comparing the drift of Laurentia and Baltica in the Proterozoic: the importance of key palaeomagnetic poles. Tectonophysics 319, 167–198.

Cawood, P.A., McCausland, P.J.A., Dunning, G.R., 2001. Opening Iapetus: constraints from the Laurentian margin in Newfoundland. Geol. Soc. Am. Bull. 113 (4), 443–453.

Cawood, P.A., Pisarevsky, S.A., 2006. Was Baltica right-way-up or upside-down in the Neoproterozoic? J. Geol. Soc., Lond. 163, 1–7.

Cawood, P.A., van Gool, J.A.M., Dunning, G.R., 1996. Geological development of eastern Humber and western Dunnage zones: Corner Brook—Glover Island region, Newfoundland. Can. J. Earth Sci. 33, 182–198.

Chase, C.G., 1985. The geological significance of the geoid. Annu. Rev. Earth Planet. Sci. 13, 97–117.

Cocks, L.R., Torsvik, T.H., 2005. Baltica from the late Precambrian to mid-Paleozoic times: the gain and loss of a terrane's identity. Earth Sci. Rev. 72, 39–66.

Cordani, U.G., D'Agrella-Filho, M.S., Brito-Neves, B.B., Trindade, R.I.F., 2003. Tearing up Rodinia: the Neoproterozoic palaeogeography of South American cratonic fragments. Terra Nova 15, 350–359.

Corsetti, F.A., Kaufman, A.J., 2003. Stratigraphic investigations of carbon isotope anomalies and Neoproterozoic ice ages in Death Valley, California. Geol. Soc. Am. Bull. 115 (8), 916–932.

Cox, R.A., Hodych, J.P., 2005. LA-ICP-MS zircon dating of the Lac Matapedia and Mt. St.-Anselme basalts, Quebec: Support for a protracted mantle plume event during the opening of Iapetus. In: Geological Association of Canada—Mineralogical Association of Canada Joint Meeting, vol. 30, Halifax, p. 38.

Dalziel, I.W.D., 1997. Neoproterozoic–Paleozoic geography and tectonics: review, hypothesis, environmental speculation. Geol. Soc. Am. Bull. 190 (1), 16–42.

Dalziel, I.W.D., Soper, N.J., 2001. Neoproterozoic extension on the Scottish promontory of Laurentia; paleogeographic and tectonic implications. J. Geol. 109, 299–317.

Dankers, P., Lapointe, P., 1981. Paleomagnetism of Lower Cambrian volcanics and a cross-cutting Cambro-Ordovician diabase dyke from Buckingham (Quebec). Can. J. Earth Sci. 18, 1174–1186.

Deutsch, E.R., 1963. Polar wandering and continental drift: an evaluation of recent evidence. In: Munyan, A.C. (Ed.), Polar Wandering and Continental Drift. Society of Economic Paleontologists and Mineralogists, Tulsa, pp. 4–46.

Dix, G.R., Salad Hersi, O., Nowlan, G.S., 2004. The Potsdam-Beekmantown Group boundary, Nepean Formation type section (Ottawa, Ontario): a crytic sequence boundary, not a conformable transition. Can. J. Earth Sci. 41, 897–902.

Doig, R., Barton, J.M., 1968. Ages of carbonatites and other igneous rocks in Quebec. Can. J. Earth Sci. 5, 1401–1407.

Dunlop, D.J., 2002. Theory and application of the Day plot (M_{rs}/M_s versus H_{cr}/H_c) 1. Theoretical curves and tests using titanomagnetite data. J. Geophys. Res. 107 (B3), 2056, doi:10.1029/2001JB000486.

Dunlop, D.J., Argyle, K.S., 1991. Separating multidomain and single-domain-like remanences in pseudo-single-domain magnetites (215–540 nm) by low-temperature demagnetization. J. Geophys. Res. 96, 2007–2017.

Dunn, W.J., Elmore, R.D., 1985. Paleomagnetic and petrographic investigations of the Taum Sauk limestone, southeast Missouri. J. Geophys. Res. 90, 11469–11483.

Easton, R.M., 2000. Metamorphism of the Canadian sheild, Ontario, Canada. II. Proterozoic metamorphic history. Can. Mineral. 38, 319–344.

Elston, D.P., Bressler, S.L., 1977. Paleomagnetic poles and polarity zonation from Cambrian and Devonian strata of Arizona. Earth Planet. Sci. Lett. 36, 423–433.

Evans, D.A.D., 1998. True polar wander, a supercontinental legacy. Earth Planet. Sci. Lett. 157, 1–8.

Evans, D.A.D., 2003. True polar wander and supercontinents. Tectonophysics 362, 303–320.

Everitt, R.A., Lorencak, M., Osadetz, K., 2005. Thermal history of Precambrian shield in Ontario from apatite fission track thermochronology. In: Geological Association of Canada—Mineralogical Association of Canada Joint Meeting, vol. 30, Halifax, p. 55.

Farr, M.R., Gose, W.A., 1991. Paleomagnetism of the Cambrian Moore Hollow Group, Texas: evidence for a primary magnetization carried by detrital magnetite. J. Geophys. Res. 96 (B6), 9895–9907.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. A 217, 295–305.

Geissman, J.W., Jackson, M., Harlan, S.S., Van der Voo, R., 1991. Paleomagnetism of Latest Cambrian–Early Ordovician and Latest Cretaceous–Early Tertiary rocks of the Florida Mountains, southwest New Mexico. J. Geophys. Res. 96 (B4), 6053–6071.

Golderich, P., Toomre, A., 1969. Some remarks on true polar wandering. J. Geophys. Res. 74, 2555–2567.

Gradstein, F.M., Ogg, J.G., Smith, A.G., 2004. A Geologic Time Scale 2004. Cambridge University Press, Cambridge.

Greig, S.C., 1968. The geology of Rigaud Mountain, Quebec. M.Sc. Thesis. McGill University, Montreal.

Grotzinger, J.P., Bowring, S.A., Saylor, B.Z., Kaufman, A.J., 1995. Biostratigraphic and geochronologic constraints on early animal evolution. Science 270, 598–604.

Gurnis, M., Torsvik, T.H., 1994. Rapid drift of large continents during the late Precambrian and Paleozoic: Paleomagnetic constraints and dynamic models. Geology 22, 1023–1026.

Halverson, G.P., Hoffman, P.F., Schrag, D.P., Maloof, A.C., Price, A.H.N., 2005. Toward a Neoproterozoic composite carbon-isotope record. Geol. Soc. Am. Bull. 117 (9), 1181–1207.

Harland, W.B., Gayer, R.A., 1972. The Arctic Caledonides and earlier oceans. Geol. Mag. 109, 289–314.

Hartz, E.H., Torsvik, T.H., 2002. Baltica upside down: a new plate tectonic model for Rodinia and the Iapetus Ocean. Geology 30, 255–258.

Higgins, M.D., 1985. Geochemical evolution of the Chatham-Grenville stock, Quebec. Can. J. Earth Sci. 22, 872–880.

Higgins, M.D., van Breemen, O., 1998. The age of the Sept Iles layered mafic intrusion, Canada: implications for the Late Neoproterozoic/Cambrian history of southeastern Canada. J. Geol. 106, 421–431.

Hodych, J.P., Cox, R.A., Kosler, J., 2004. An equatorial Laurentia at 550 Ma confirmed by Grenvillian inherited zircons dated by LAM ICP-MS in the Skinner Cove volcanics of western Newfoundland: implications for inertial interchange true polar wander. Precambr. Res. 129, 93–113.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwanaland inside out? Science 252, 1409–1412.

Hoffman, P.F., Schrag, D.P., 2002. The snowball Earth hypothesis: testing the limits of global change. Terra Nova 14, 129–155.

Hogan, J.P., Gilbert, M.C., Price, J.D., 2000. Crystallisation of fine- and coarse-grained A-type granite sheets of the southern Oklahoma aulacogen, U.S.A. Trans. R. Soc. Edinburgh: Earth Sci. 91, 139–150.

Iglesia-Llanos, M.P., Tait, J., Popov, V., Ablamasova, A., 2005. Palaeomagnetic data from Ediacaran (Vendian) sediments of the Arkhangelsk region, NW Russia: an alternative apparent polar wander path of Baltica for the Late Proterozoic–Early Palaeozoic. Earth Planet. Sci. Lett. 240, 732–747.

Kamo, S.L., Heaman, L.M., Lumbers, S.B., 1989. Birthdate for the Iapetus Ocean? A precise U–Pb zircon and baddeleyite age for the Long Range dykes, southeast Labrador. Geology 17, 602–605.

Kamo, S.L., Gower, C.F., 1994. Note: U–Pb baddeleyite dating clarifies age of characteristic paleomagnetic remanence of Long Range dykes, southeastern Labrador. Atlantic Geol. 30, 259–262.

Kamo, S.L., Krogh, T.E., Kumarapeli, P.S., 1995. Age of the Grenville dyke swarm, Ontario-Quebec: implications for the timing of Iapetus rifting. Can. J. Earth Sci. 32, 273–280.

Kilner, B., MacNiocaill, C., Brasier, M., 2005. Low-latitude glaciation in the Neoproterozoic of Oman. Geology 33 (5), 413–416.

Kirschvink, J.L., 1980. The least squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astronom. Soc. 62, 699–718.

Kirschvink, J.L., Ripperdan, R.L., Evans, D.A.D., 1997. Evidence for a large-scale reorganization of Early Cambrian continental landmasses by inertial interchange true polar wander. Science 277, 541–545.

Knoll, A.H., Walter, M.R., Narbonne, G.M., Christie-Blick, N., 2004. A new period for the Geologic Time Scale. Science 305, 621–622.

Kravchinsky, V.A., Konstantinov, K.M., Cogne, J.-P., 2001. Palaeomagnetic study of Vendian and Early Cambrian rocks of South Siberia and Central Mongolia: was the Siberian platform assembled at this time? Precambr. Res. 110, 61–92.

Kumarapeli, P.S., 1993. A plume-generated segment of the rifted margin of Laurentia as seen through a completed Wilson Cycle. Tectonophysics 219, 47–55.

Kumarapeli, P.S., Dunning, G.R., Pintson, H., Shaver, J., 1989. Geochemistry and U–Pb zircon age of comenditic metafelsites of the Tibbit Hill Formation, Quebec Appalachians. Can. J. Earth Sci. 26, 1374–1383.

Kumarapeli, P.S., Saull, V.A., 1966. The St. Lawrence valley system: a North American equivalent of the East African rift valley system. Can. J. Earth Sci. 3, 639–658.

Lafleur, J., Hogarth, D.D., 1981. Cambro-Proterozoic volcanism near Buckingham, Quebec. Can. J. Earth Sci. 18, 1817–1823.

Lawver, L., Scotese, C.R., 1987. A revised reconstruction of Gondwanaland. In: McKenzie, G.D. (Ed.), Gondwana Six: Structure, Tectonics and Geophysics. American Geophysical Union Monograph, pp. 17–23.

Loucks, V., Elmore, R.D., 1986. Absolute dating of dedolomitization and the origin of a magnetization in the Cambrian Morgan Creek limestone, central Texas. Geol. Soc. Am. Bull. 97, 486–496.

Lukert, M.T., Banks, P.O., 1984. Geology and age of the Robertson River pluton. In: Bartholomew, M.J. (Ed.), The Grenville Event in the Appalachians and Related Topics. Geological Society of America, pp. 161–166.

Malka, E., Stevenson, R.K., David, J., 2000. Sm–Nd geochemistry and U–Pb geochronology of the Mont Rigaud stock, Quebec, Canada: a late magmatic event associated with the formation of the Iapetus rift. J. Geol. 108, 569–583.

McCausland, P.J.A., Hodych, J.P., 1998. Paleomagnetism of the 550 Ma Skinner Cove volcanics of western Newfoundland and the opening of the Iapetus Ocean. Earth Planet. Sci. Lett. 163, 15–29.

McCausland, P.J.A., Van der Voo, R., 2004. Rapid, coherent apparent polar wander in the terminal Neoproterozoic–Early Cambrian? EOS: Trans. Am. Geophys. Union 85 (47), Abstract U33A-0026.

McDougall, I., Harrison, T.M., 1999. Geochronology and Thermochronology by the $^{40}Ar/^{39}Ar$ Method. Oxford University Press, Oxford, 269 pp.

McFadden, P.L., Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions. Geophys. J. R. Astronom. Soc. 67, 19–33.

McFadden, P.L., McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in paleomagnetism. Earth Planet. Sci. Lett. 87, 161–172.

Meert, J.G., 1999. A paleomagnetic analysis of Cambrian true polar wander. Earth Planet. Sci. Lett. 168, 131–144.

Meert, J.G., Lieberman, B.S., 2004. A paleomagnetic and paleobiogeographical perspective on latest Neoproterozoic and early Cambrian tectonic events. J. Geol. Soc., Lond. 161, 477–487.

Meert, J.G., Powell, C.M., 2001. Introduction to the special volume on the assembly and breakup of Rodinia. Precambr. Res. 110, 1–8.

Meert, J.G., Torsvik, T.H., 2003. The making and unmaking of a supercontinent: Rodinia revisited. Tectonophysics 375, 261–288.

Meert, J.G., Van der Voo, R., 1996. Paleomagnetic and $^{40}Ar/^{39}Ar$ study of the Sinyai dolerite, Kenya: implications for Gondwana assembly. J. Geol. 104, 131–142.

Meert, J.G., Van der Voo, R., Payne, T.W., 1994. Paleomagnetism of the Catoctin volcanic province: a new Vendian Cambrian apparent polar wander path for North America. J. Geophys. Res. 99, 4625–4641.

Meert, J.G., et al., 1993. A plate tectonic speed limit? Nature 363, 217–218.

Miller, B.V., Barr, S.M., 2004. Metamorphosed gabbroic dikes related to the opening of Iapetus Ocean at the St. Lawrence promontory: Blair River inlier, Nova Scotia, Canada. J. Geol. 112, 277–288.

Molina-Garza, R., Geissman, J.W., Lucas, S., Van der Voo, R., 1996. Paleomagnetism and magnetostratigraphy of Triassic strata in the Sangre de Cristo Mountains and Tucumcari basin, New Mexico, U.S.A. Geophys. J. Int. 124, 935–953.

Murphy, J.B., Pisarevsky, S.A., Nance, R.D., Keppie, J.D., 2004. Neoproterozoic-Early Paleozoic evolution of peri-Gondwanan terranes: Implications for Laurentia–Gondwana connections. Int. J. Earth Sci. (Geol. Rundsch.) 93, 659–682.

Murthy, G.S., 1971. The paleomagnetism of diabase dykes from the Grenville Province. Can. J. Earth Sci. 8 (802–812).

Murthy, G.S., Gower, C.F., Tubrett, M., Patzold, R., 1992. Paleomagnetism of Eocambrian Long Range dykes and Double Mer Formation from Labrador, Canada. Can. J. Earth Sci. 29, 1224–1234.

Nawrocki, J., Boguckij, A., Katanas, V., 2004. New late Vendian palaeogeography of Baltica and TESZ. Geol. Quart. 48, 309–316.

Nick, K.E., Elmore, R.D., 1990. Paleomagnetism of the Cambrian Royer dolomite and Pennsylvaniam Collings Ranch conglomerate, southern Oklahoma: an early Paleozoic magnetization and nonpervasive remagnetization by weathering. Geol. Soc. Am. Bull. 102, 1517–1525.

Park, J.K., 1995. Paleomagnetism of the late Neoproterozoic Blueflower and Risky formations of the northern Cordillera, Canada. Can. J. Earth Sci. 32, 718–729.

Pavlov, V., Gallet, Y., 2001. Middle Cambrian high magnetic reversal frequency (Kulumbe River section, northwestern Siberia) and reversal behaviour during the Early Palaeozoic. Earth Planet. Sci. Lett. 185, 173–183.

Philpotts, A.R., 1976. Canton de Grenville (partie SE). Ministere des richesses naturelles, Quebec. Rapport Geol., 156.

Pisarevsky, S.A., 2005. New edition of the Global Paleomagnetic Database. EOS Trans. 86, 170.

Pisarevsky, S.A., Komissarova, R.A., Khramov, A.N., 2001. Reply to comment by J. Meert and R. Van der Voo on "New paleomagnetic result from the Vendian red sediments in Cisbaialka and the problem of Siberia and Laurentia in the Vendian". Geophys. J. Int. 146, 871–873.

Pisarevsky, S.A., Natapov, L.M., 2003. Siberia and Rodinia. Tectonophysics 375, 221–245.

Pisarevsky, S.A., Gurevich, E.L., Khramov, A.N., 1997. Palaeomagnetism of the Lower Cambrian sediments from the Olenek river section (northern Siberia): paleopoles and a problem of magnetic polarity in Early Cambrian. Geophys. Jour. Inter. 130, 746–756.

Popov, V., Iosifidi, A., Khramov, A.N., Tait, J., Bachtadse, V., 2002. Paleomagnetism of Upper Vendian sediments from the Winter Coast, White Sea region, Russia: Implications for the paleogeography of Baltica during Neoproterozoic times. J. Geophys. Res. 107 (B11), 2315 (doi:10.1029/2001JB001607).

Powell, C.M., 1995. Comment on paper by G.M. Young, "Are Neoproterozoic glacial deposits preserved on the margins of Laurentia related to the fragmentation of two supercontinents?". Geology 23, 1053–1054.

Puffer, J.H., 2002. A late Neoproterozoic eastern Laurentian superplume: location, size, chemical composition, and environmental impact. Am. J. Sci. 302, 1–27.

Rao, K.V., Deutsch, E.R., 1976. Paleomagnetism of the Lower Cambrian Bradore sandstones, and the rotation of Newfoundland. Tectonophysics 33, 337–367.

Roggenthen, W.M., Fischer, J.F., Napoleone, G., Fischer, A.G., 1981. Paleomagnetism and age of mafic plutons, Wichita Mountains, Oklahoma. Geophys. Res. Lett. 8, 133–136.

Sears, J.W., Price, R.A., 2003. Tightening the Siberian connection to western Laurentia. Geol. Soc. Am. Bull. 115 (8), 943–953.

Samson, S.D., Alexander, E.C., 1987. Calibration of the interlaboratory ^{40}Ar–^{39}Ar dating standard, MMhb-1. Chem. Geol. 66, 27–34.

Sánchez-Bettucci, L., Rapalini, A.E., 2002. Paleomagnetism of the Sierra de Las Animas Complex, southern Uruguay: its implications in the assembly of western Gondwana. Precam. Res. 118, 243–265.

Smethurst, M.A., Khramov, A.N., Torsvik, T.H., 1998. The Neoproterozoic and Palaeozoic palaeomagnetic data for the Siberian Platform: From Rodinia to Pangea. Earth Sci. Rev. 43, 1–24.

Spall, H., 1968. Paleomagnetism of basement granites of southern Oklahoma and its implications, progress report. Oklahoma Geol. Notes 28, 65–80.

Streepey, M.M., Lithgow-Bertelloni, C., van der Pluijm, B.A., Essene, E.J., Magloughlin, J.F., 2004. Exhumation of a collisional orogen: a perspective from the North American Grenville Province. In: Tollo, R.P., Corriveau, L., McLelland, J., Bartholomew, M.J. (Eds.), Proterozoic Tectonic Evolution of the Grenville Orogen in North America. Geological Society of America, Boulder, pp. 391–410.

Stukas, V., Reynolds, P.H., 1974. $^{40}Ar/^{39}Ar$ dating of the Long Range dikes, Newfoundland. Earth Planet. Sci. Lett. 22, 256–266.

Su, Q., Goldberg, S.A., Fullagar, P.D., 1994. Precise U–Pb zircon ages of Neoproterozoic plutons in the southern Appalachian Blue Ridge and their implications for the initial rifting of Laurentia. Precambr. Res. 68, 81–95.

Symons, D.T.A., Chiasson, A.D., 1991. Paleomagnetism of the Callander complex and the Cambrian apparent polar wander path for North America. Can. J. Earth Sci. 28, 355–363.

Tanczyk, E.I., Lapointe, P., Morris, W.A., Schmidt, P.W., 1987. A paleomagnetic study of the layered mafic intrusion at Sept-Iles, Quebec. Can. J. Earth Sci. 24, 1431–1438.

Tohver, E., D'Agrella-Filho, M.S., Trindade, R.I.F., 2006a. Paleomagnetic record of Africa and South America for the 1200–500 Ma interval, and evaluation of Rodinia and Gondwana assemblies. Precambr. Res. 147, 193–222.

Tohver, E., Teixeira, W., van der Pluijm, B.A., Geraldes, M.C., Bettencourt, J.S., Rizzotto, G., 2006b. Restored transect across the exhumed Grenville orogen of Laurentia and Amazonia, with implications for crustal architecture. Geology 34 (8), 669–672 (doi:10.1130/G22534.1).

Tollo, R.P., Aleinikoff, J.N., Bartholomew, M.J., Rankin, D.W., 2004. Neoproterozoic A-type granitoids of the central and southern Appalachians: intraplate magmatism associated with episodic rifting of the Rodinian supercontinent. Precambr. Res. 128, 3–38.

Torsvik, T.H., Rehnstrom, E.F., 2001. Cambrian palaeomagnetic data from Baltica: implications for true polar wander and Cambrian palaeogeography. J. Geol. Soc., Lond. 158, 321–329.

Trindade, R.I.F., D'Agrella-Filho, M.S., Epof, I., Brito-Neves, B.B., 2006. Paleomagnetism of Early Cambrian Itabaiana mafic dikes (NE Brazil) and the final assembly of Gondwana. Earth Planet. Sci. Lett. 244, 361–367.

Trindade, R.I.F., Font, E., D'Agrella-Filho, M.S., Nogueira, A.C.R., Riccomini, C., 2003. Low-latitude and multiple geomagnetic reversals in the Neoproterozoic Puga cap carbonate, Amazon craton. Terra Nova 15, 441–446.

Van Alstine, D.R., Gillett, S.L., 1979. Paleomagnetism of the upper Precambrian sedimentary rocks from the Desert Range, Nevada. J. Geophys. Res. 84, 4490–4500.

van Berkel, J.T., Currie, K.L., 1988. Geology of the Puddle Pond (12A/5) and Little Grand Lake (12A/12) map areas, southwestern Newfoundland. Newfoundland Department of Mines, Mineral Development Division Report 88-1, 99–107.

Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.

Van der Voo, R., 1993. Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans. Cambridge University Press, Cambridge, 411 pp.

Van der Voo, R., French, R.B., Williams, D.W., 1976. Paleomagnetism of the Wilberns Formation (Texas) and the Late Cambrian paleomagnetic field for North America. J. Geophys. Res. 81, 5633–5638.

Vincenz, S.A., Yaskawa, K., Ade-Hall, J.M., 1975. Origin of the magnetization of the Wichita Mountains granites, Oklahoma. Geophys. J. R. Astronom. Soc. 42, 21–48.

Waggoner, B., 1999. Biogeographic analyses of the Ediacara biota: a conflict with paleotectonic reconstructions. Paleobiology 25 (4), 440–458.

Waldron, J.W.F., van Staal, C.R., 2001. Taconian orgeny and the accretion of the Dashwoods block: a peri-Laurentian microcontinent in the Iapetus Ocean. Geology 29 (9), 811–814.

Walsh, G.J., Aleinikoff, J.N., 1999. U–Pb zircon age of metafelsite from the Pinney Hollow Formation: implications for the development of the Vermont Appalachians. Am. J. Sci. 299, 157–170.

Williams, H., Gillespie, R.T., van Breemen, O., 1985. A late Precambrian rift-related igneous suite in western Newfoundland. Can. J. Earth Sci. 22, 1727–1735.

Williams, H., Hiscott, R.N., 1987. Definition of the rift-drift transition in western Newfoundland. Geology 15, 1044–1047.

Wright, J.E., Hogan, J.P., Gilbert, M.C., 1996. The southern Oklahoma aulacogen: Not just another B.L.I.P. EOS: Trans. Am. Geophys. Union 77 (46), F845.

Yu, Y., Dunlop, D.J., Ozdemir, O., 2003. On the resolution of multivectorial remanences. Earth Planet. Sci. Lett. 208, 13–26.

Zhang, S., et al., 2005. U–Pb sensitive high-resolution ion microprobe ages from the Doushantuo Formation in south China: constraints on late Neoproterozoic glaciations. Geology 33, 473–476.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, The Netherlands, pp. 254–286.